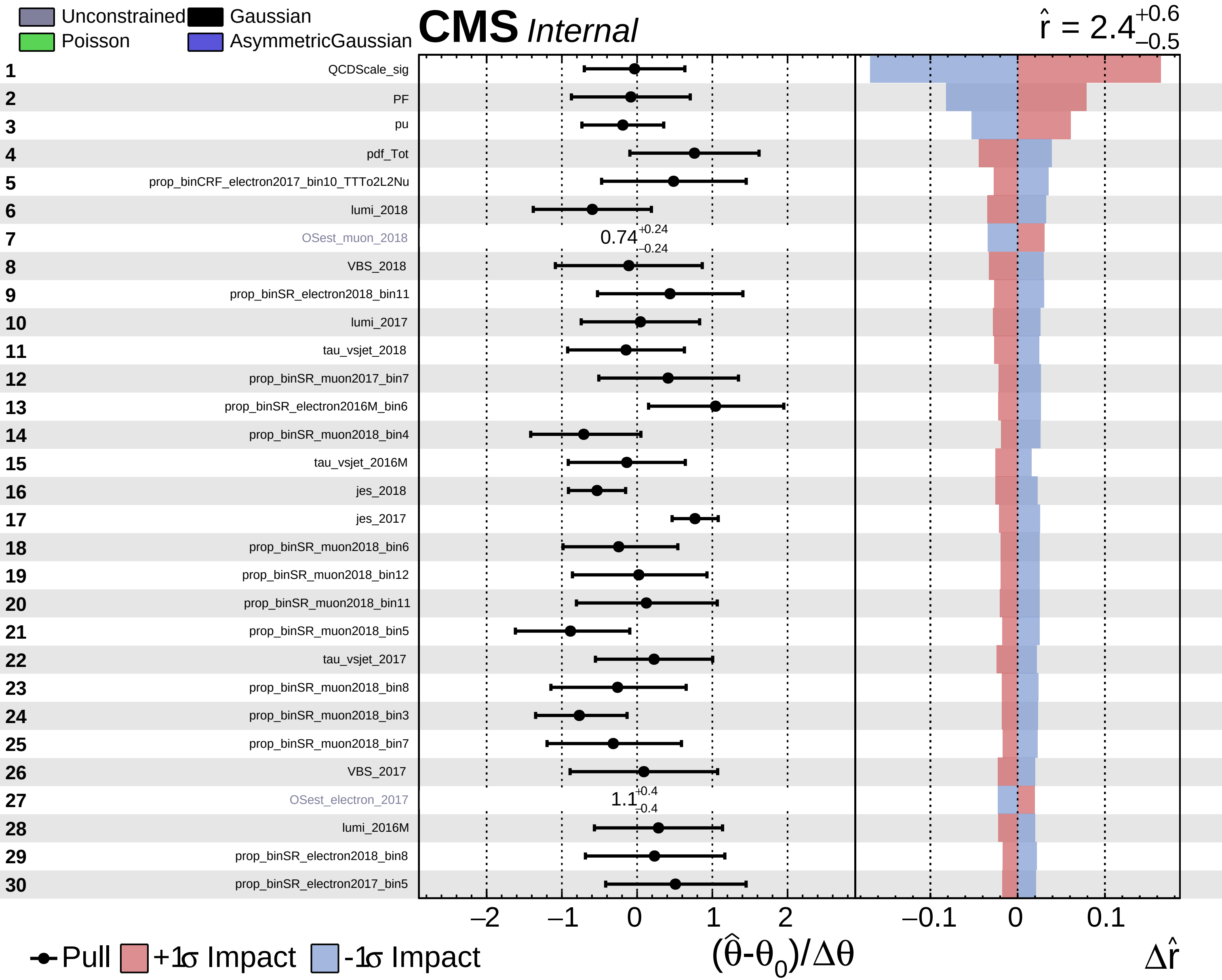
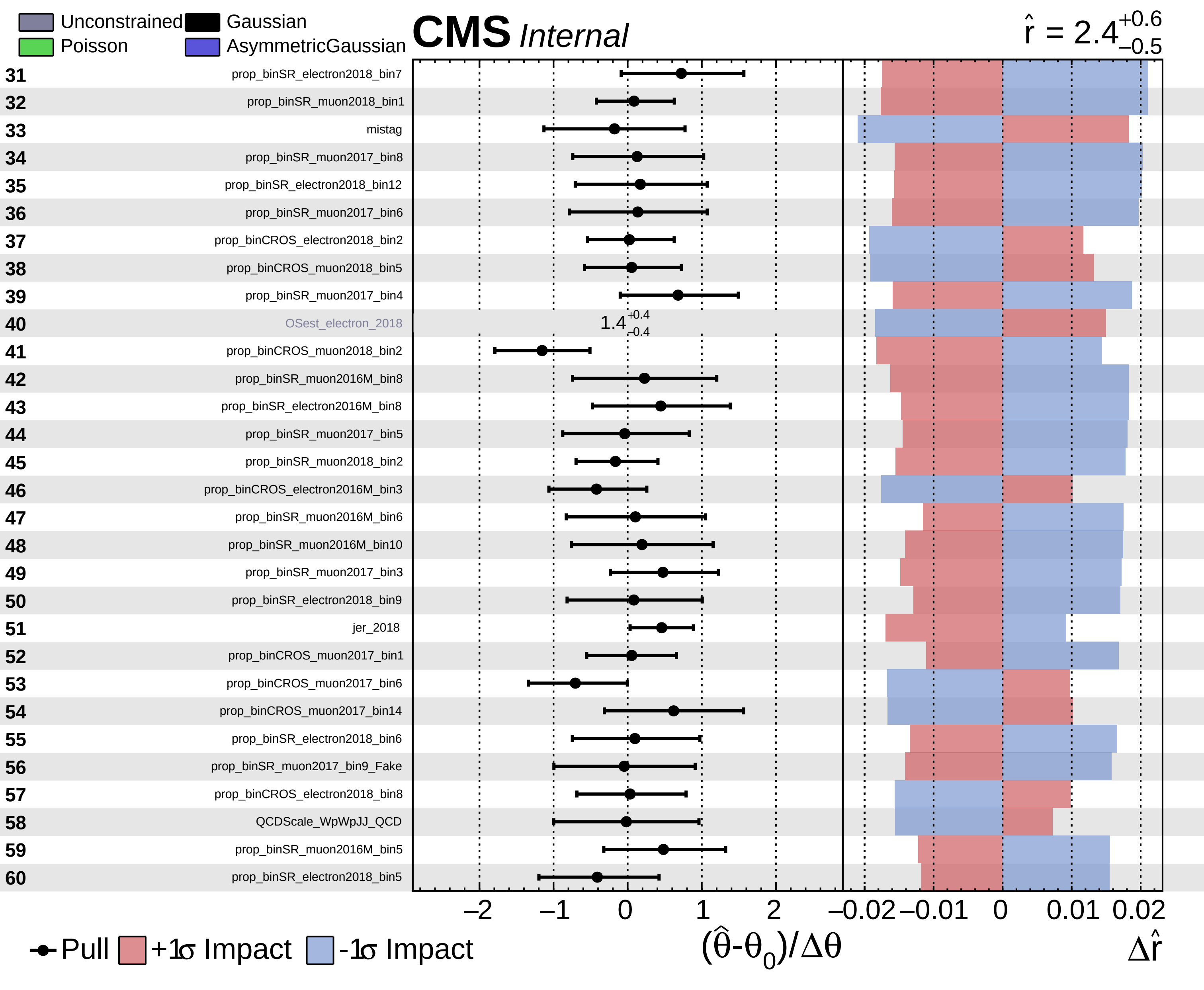
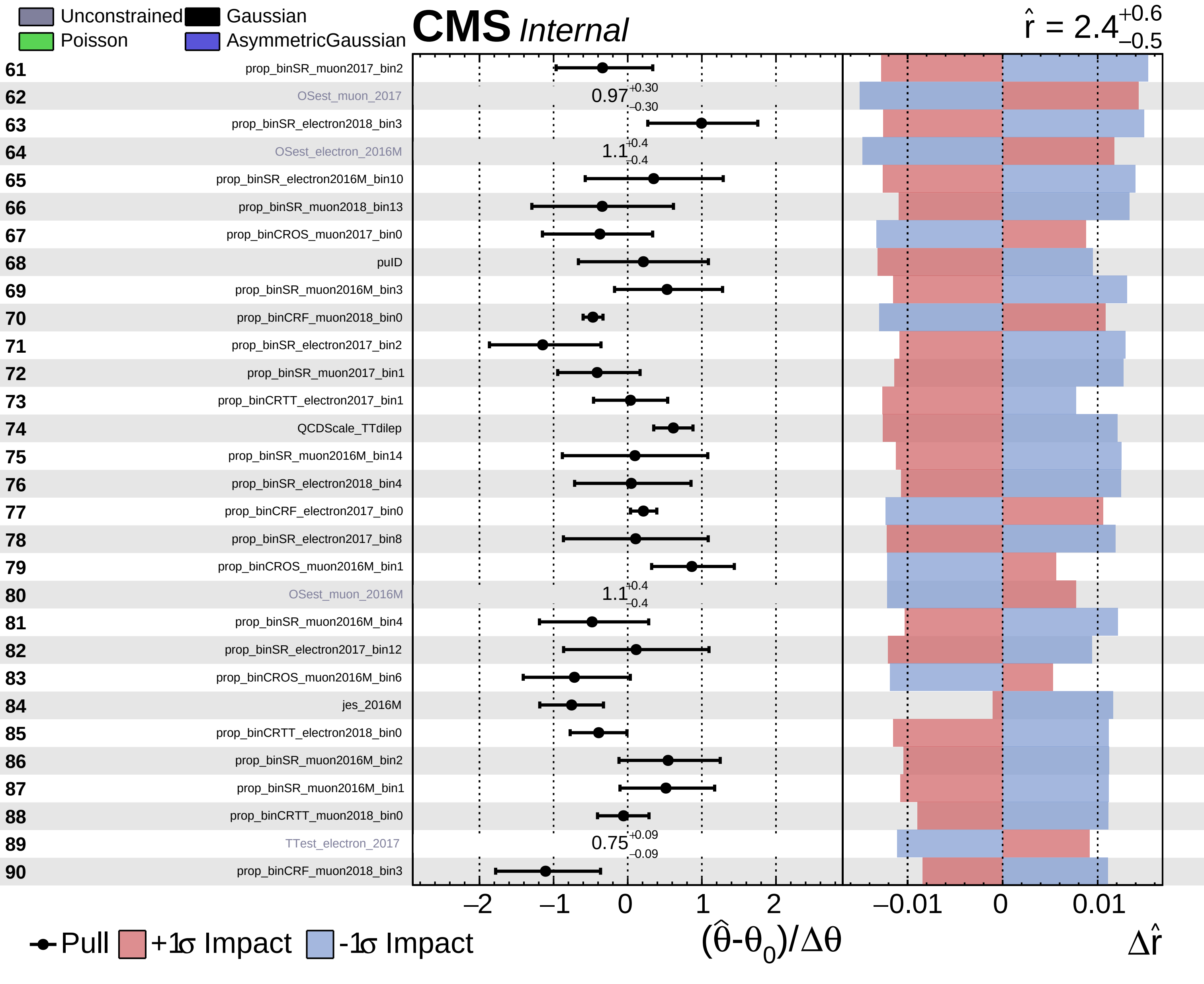


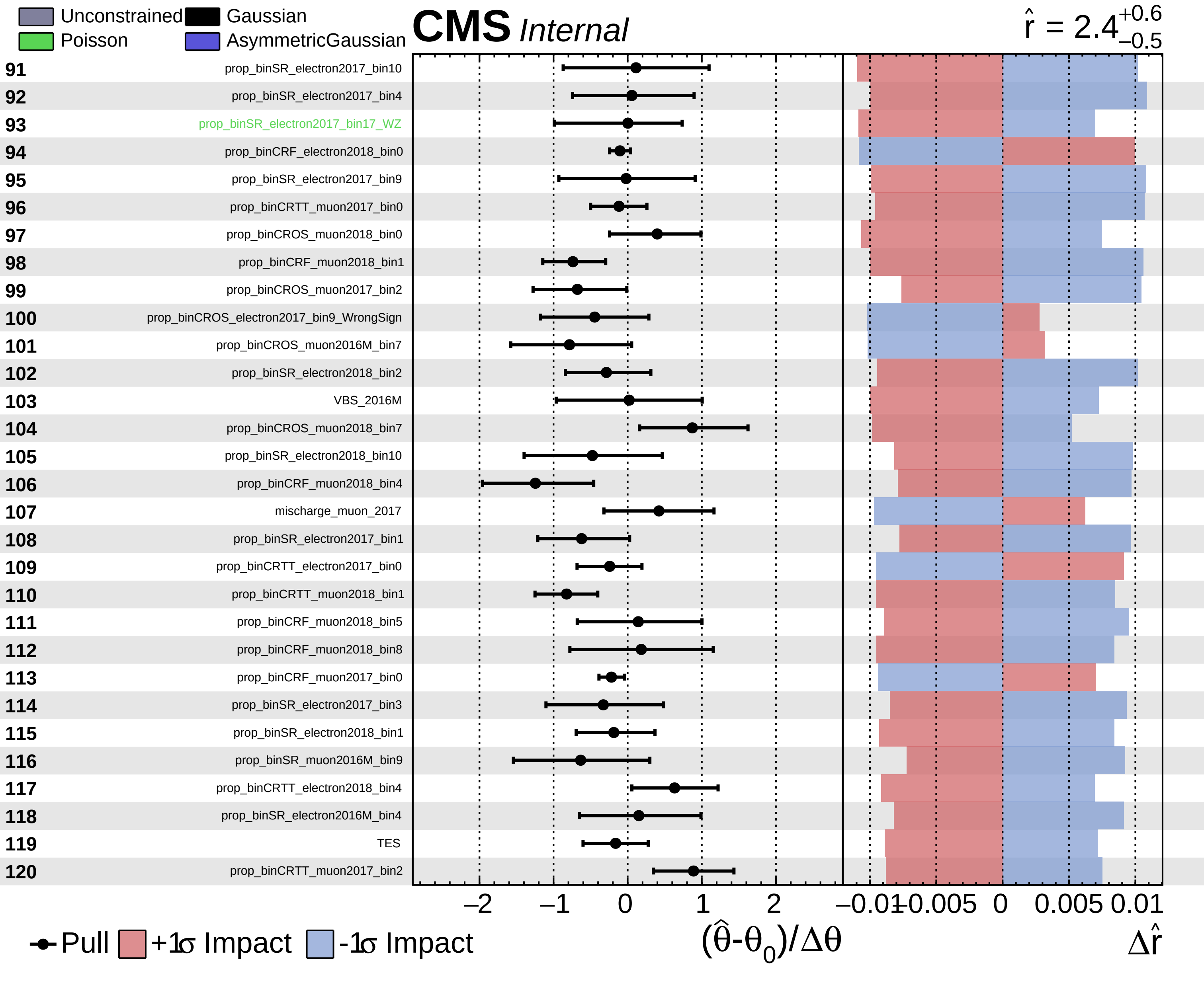
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 2.4^{+0.6}_{-0.5}$
1 QCDScale_sig
2 PF
3 pu
4 pdf_Tot
5 prop_binCRF_electron2017_bin10_TTTo2L2Nu
6 lumi_2018
7 OSest_muon_2018
$0.74^{+0.24}_{-0.24}$
8 VBS_2018
9 prop_binSR_electron2018_bin11
10 lumi_2017
11 tau_vsjet_2018
12 prop_binSR_muon2017_bin7
13 prop_binSR_electron2016M_bin6
14 prop_binSR_muon2018_bin4
15 tau_vsjet_2016M
16 jes_2018
17 jes_2017
18 prop_binSR_muon2018_bin6
19 prop_binSR_muon2018_bin12
20 prop_binSR_muon2018_bin11
21 prop_binSR_muon2018_bin5
22 tau_vsjet_2017
23 prop_binSR_muon2018_bin8
24 prop_binSR_muon2018_bin3
25 prop_binSR_muon2018_bin7
26 VBS_2017
27 OSest_electron_2017
$1.1^{+0.4}_{-0.4}$
28 lumi_2016M
29 prop_binSR_electron2018_bin8
30 prop_binSR_electron2017_bin5
−2 −1 0 1 2
−0.1 0 0.1
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$



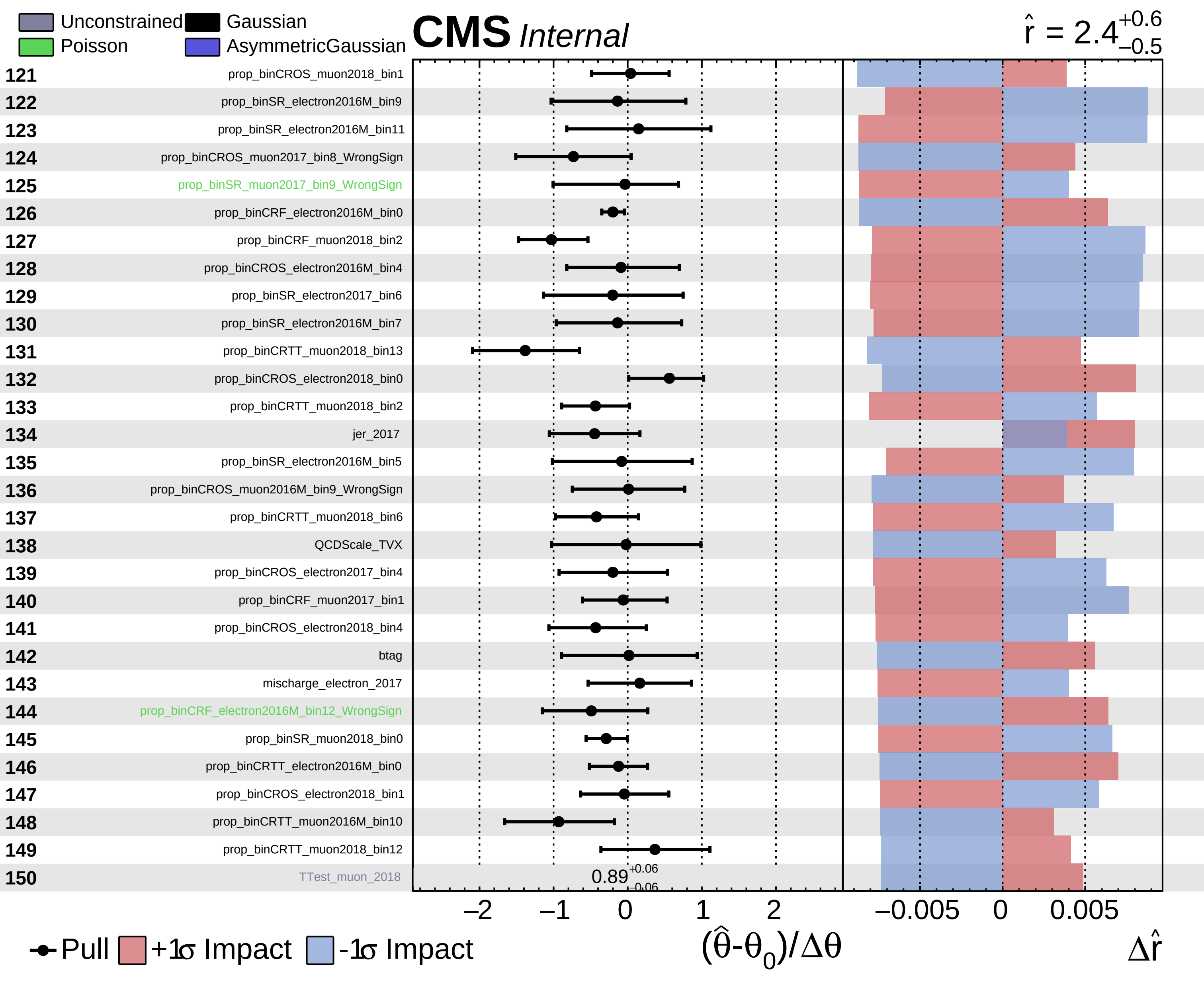
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 2.4^{+0.6}_{-0.5}$
31 prop_binSR_electron2018_bin7
32 prop_binSR_muon2018_bin1
33 mistag
34 prop_binSR_muon2017_bin8
35 prop_binSR_electron2018_bin12
36 prop_binSR_muon2017_bin6
37 prop_binCROS_electron2018_bin2
38 prop_binCROS_muon2018_bin5
39 prop_binSR_muon2017_bin4
40 OSest_electron_2018
$1.4^{+0.4}_{-0.4}$
41 prop_binCROS_muon2018_bin2
42 prop_binSR_muon2016M_bin8
43 prop_binSR_electron2016M_bin8
44 prop_binSR_muon2017_bin5
45 prop_binSR_muon2018_bin2
46 prop_binCROS_electron2016M_bin3
47 prop_binSR_muon2016M_bin6
48 prop_binSR_muon2016M_bin10
49 prop_binSR_muon2017_bin3
50 prop_binSR_electron2018_bin9
51 jer_2018
52 prop_binCROS_muon2017_bin1
53 prop_binCROS_muon2017_bin6
54 prop_binCROS_muon2017_bin14
55 prop_binSR_electron2018_bin6
56 prop_binSR_muon2017_bin9_Fake
57 prop_binCROS_electron2018_bin8
58 QCDScale_WpWpJJ_QCD
59 prop_binSR_muon2016M_bin5
60 prop_binSR_electron2018_bin5
−2
−1
0
1
2
−0.02
−0.01
0
0.01
0.02
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$



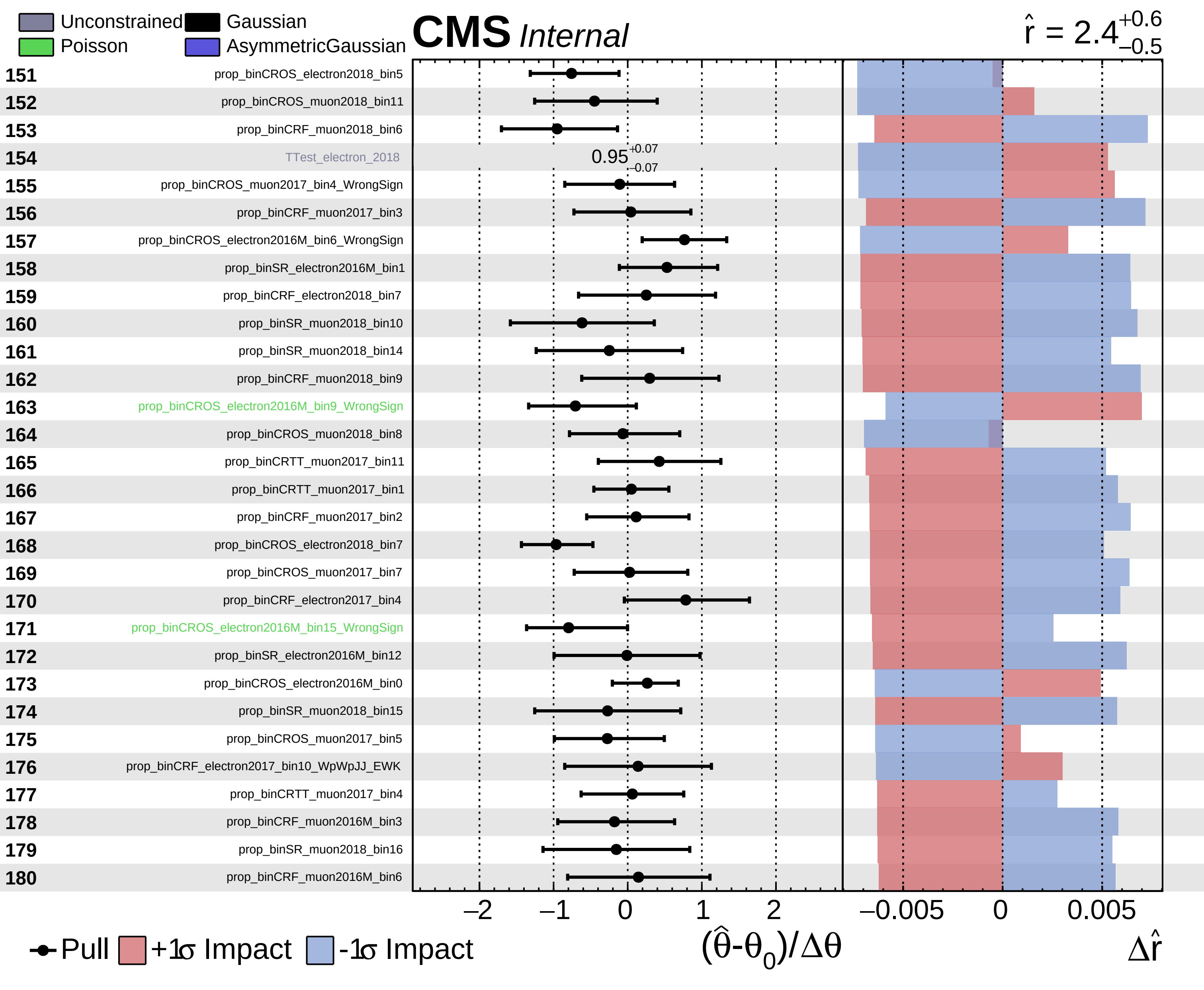
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
r̂ = 2.4 +0.6 −0.5
61 prop_binSR_muon2017_bin2
62 OSest_muon_2017 0.97 +0.30 −0.30
63 prop_binSR_electron2018_bin3
64 OSest_electron_2016M 1.1 +0.4 −0.4
65 prop_binSR_electron2016M_bin10
66 prop_binSR_muon2018_bin13
67 prop_binCROS_muon2017_bin0
68 puID
69 prop_binSR_muon2016M_bin3
70 prop_binCRF_muon2018_bin0
71 prop_binSR_electron2017_bin2
72 prop_binSR_muon2017_bin1
73 prop_binCRTT_electron2017_bin1
74 QCDScale_TTdilep
75 prop_binSR_muon2016M_bin14
76 prop_binSR_electron2018_bin4
77 prop_binCRF_electron2017_bin0
78 prop_binSR_electron2017_bin8
79 prop_binCROS_muon2016M_bin1
80 OSest_muon_2016M 1.1 +0.4 −0.4
81 prop_binSR_muon2016M_bin4
82 prop_binSR_electron2017_bin12
83 prop_binCROS_muon2016M_bin6
84 jes_2016M
85 prop_binCRTT_electron2018_bin0
86 prop_binSR_muon2016M_bin2
87 prop_binSR_muon2016M_bin1
88 prop_binCRTT_muon2018_bin0
89 TTest_electron_2017 0.75 +0.09 −0.09
90 prop_binCRF_muon2018_bin3
−2 −1 0 1 2
−0.01 0 0.01
Pull
+1σ Impact
-1σ Impact
(θ̂-θ0)/Δθ
Δr̂



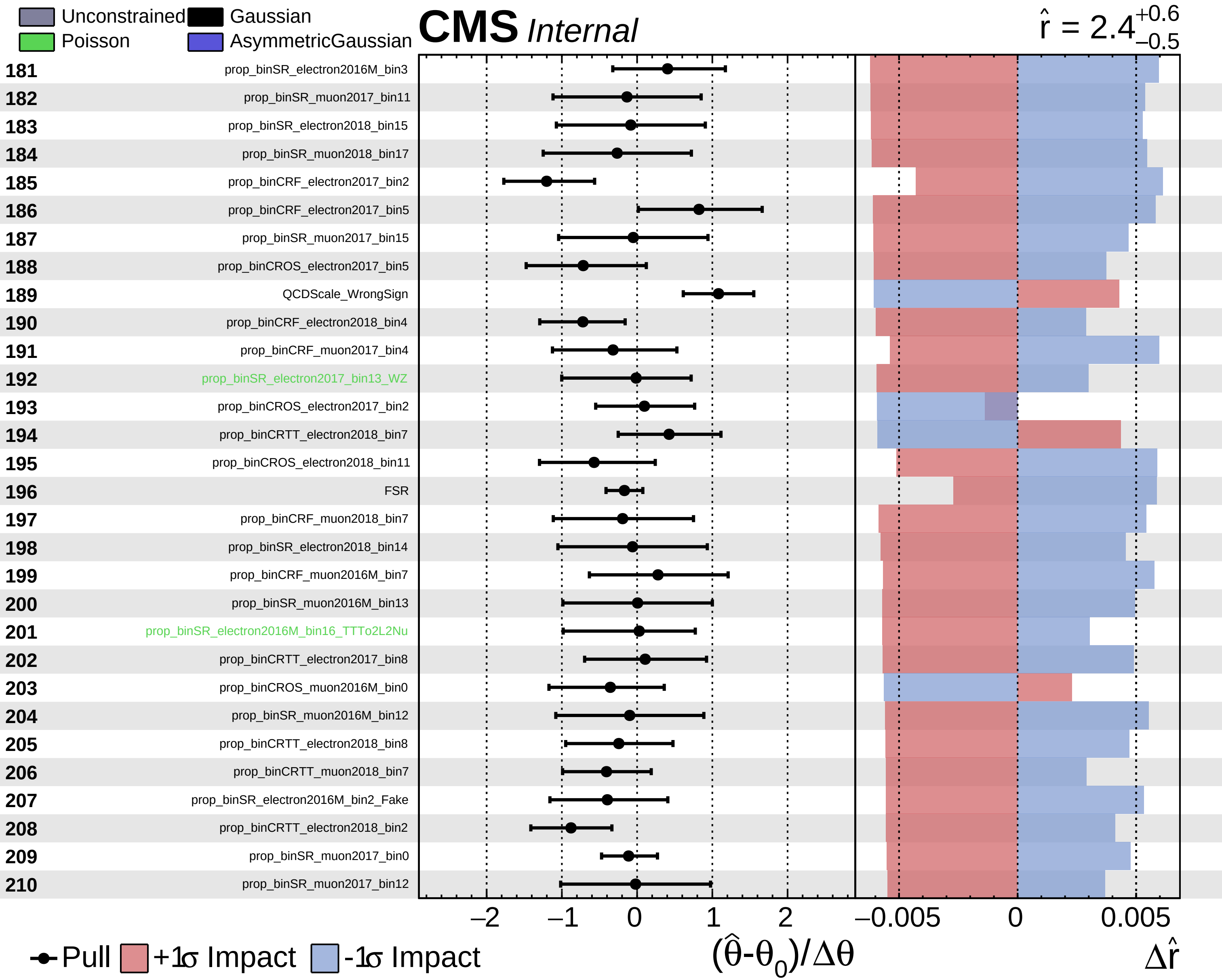
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 2.4^{+0.6}_{-0.5}$
91 prop_binSR_electron2017_bin10
92 prop_binSR_electron2017_bin4
93 prop_binSR_electron2017_bin17_WZ
94 prop_binCRF_electron2018_bin0
95 prop_binSR_electron2017_bin9
96 prop_binCRTT_muon2017_bin0
97 prop_binCROS_muon2018_bin0
98 prop_binCRF_muon2018_bin1
99 prop_binCROS_muon2017_bin2
100 prop_binCROS_electron2017_bin9_WrongSign
101 prop_binCROS_muon2016M_bin7
102 prop_binSR_electron2018_bin2
103 VBS_2016M
104 prop_binCROS_muon2018_bin7
105 prop_binSR_electron2018_bin10
106 prop_binCRF_muon2018_bin4
107 mischarge_muon_2017
108 prop_binSR_electron2017_bin1
109 prop_binCRTT_electron2017_bin0
110 prop_binCRTT_muon2018_bin1
111 prop_binCRF_muon2018_bin5
112 prop_binCRF_muon2018_bin8
113 prop_binCRF_muon2017_bin0
114 prop_binSR_electron2017_bin3
115 prop_binSR_electron2018_bin1
116 prop_binSR_muon2016M_bin9
117 prop_binCRTT_electron2018_bin4
118 prop_binSR_electron2016M_bin4
119 TES
120 prop_binCRTT_muon2017_bin2
−2 −1 0 1 2
−0.01 −0.005 0 0.005 0.01
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$



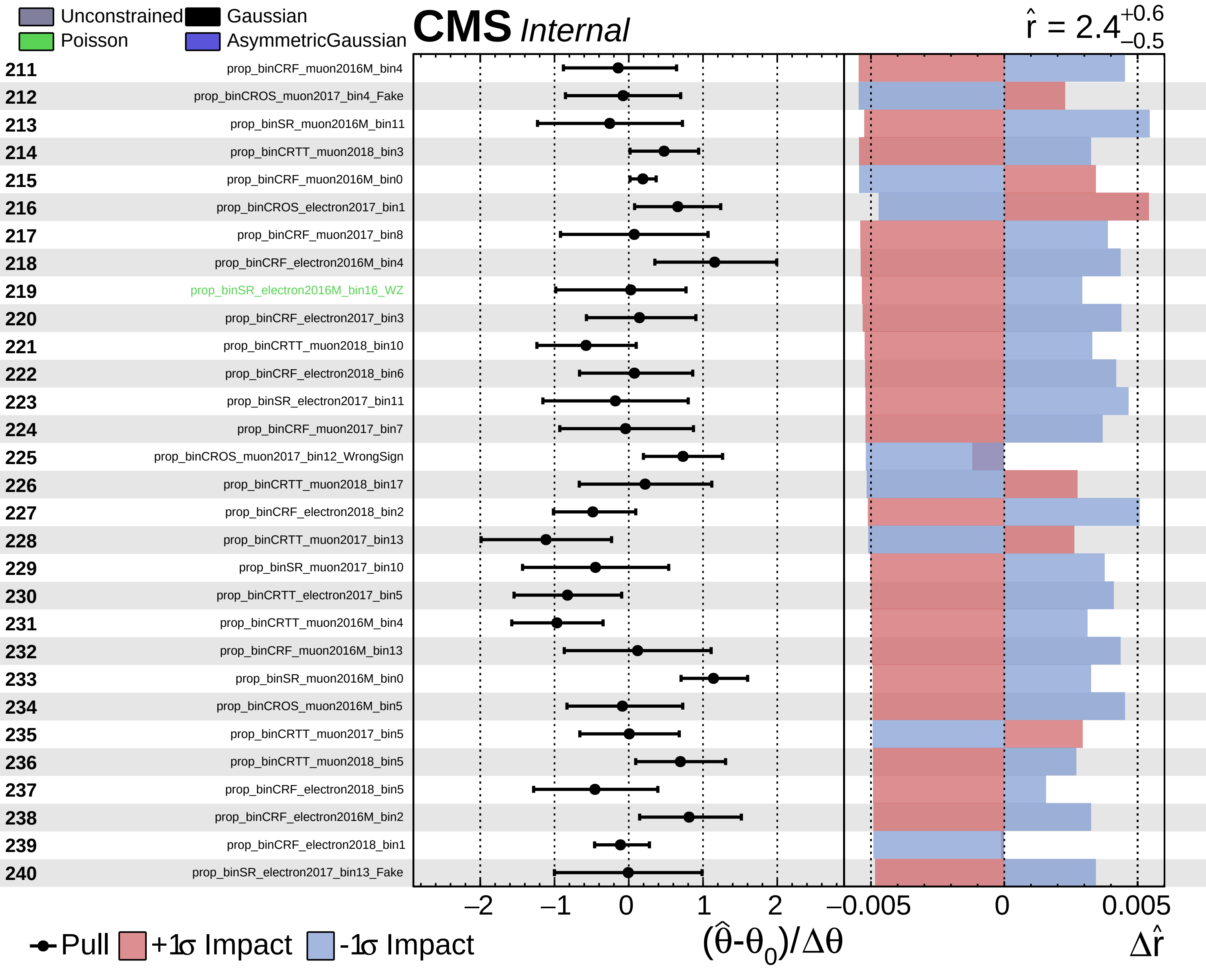
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 2.4^{+0.6}_{-0.5}$
121 prop_binCROS_muon2018_bin1
122 prop_binSR_electron2016M_bin9
123 prop_binSR_electron2016M_bin11
124 prop_binCROS_muon2017_bin8_WrongSign
125 prop_binSR_muon2017_bin9_WrongSign
126 prop_binCRF_electron2016M_bin0
127 prop_binCRF_muon2018_bin2
128 prop_binCROS_electron2016M_bin4
129 prop_binSR_electron2017_bin6
130 prop_binSR_electron2016M_bin7
131 prop_binCRTT_muon2018_bin13
132 prop_binCROS_electron2018_bin0
133 prop_binCRTT_muon2018_bin2
134 jer_2017
135 prop_binSR_electron2016M_bin5
136 prop_binCROS_muon2016M_bin9_WrongSign
137 prop_binCRTT_muon2018_bin6
138 QCDScale_TVX
139 prop_binCROS_electron2017_bin4
140 prop_binCRF_muon2017_bin1
141 prop_binCROS_electron2018_bin4
142 btag
143 mischarge_electron_2017
144 prop_binCRF_electron2016M_bin12_WrongSign
145 prop_binSR_muon2018_bin0
146 prop_binCRTT_electron2016M_bin0
147 prop_binCROS_electron2018_bin1
148 prop_binCRTT_muon2016M_bin10
149 prop_binCRTT_muon2018_bin12
150 TTest_muon_2018
$0.89^{+0.06}_{-0.06}$
−2 −1 0 1 2
−0.005 0 0.005
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$

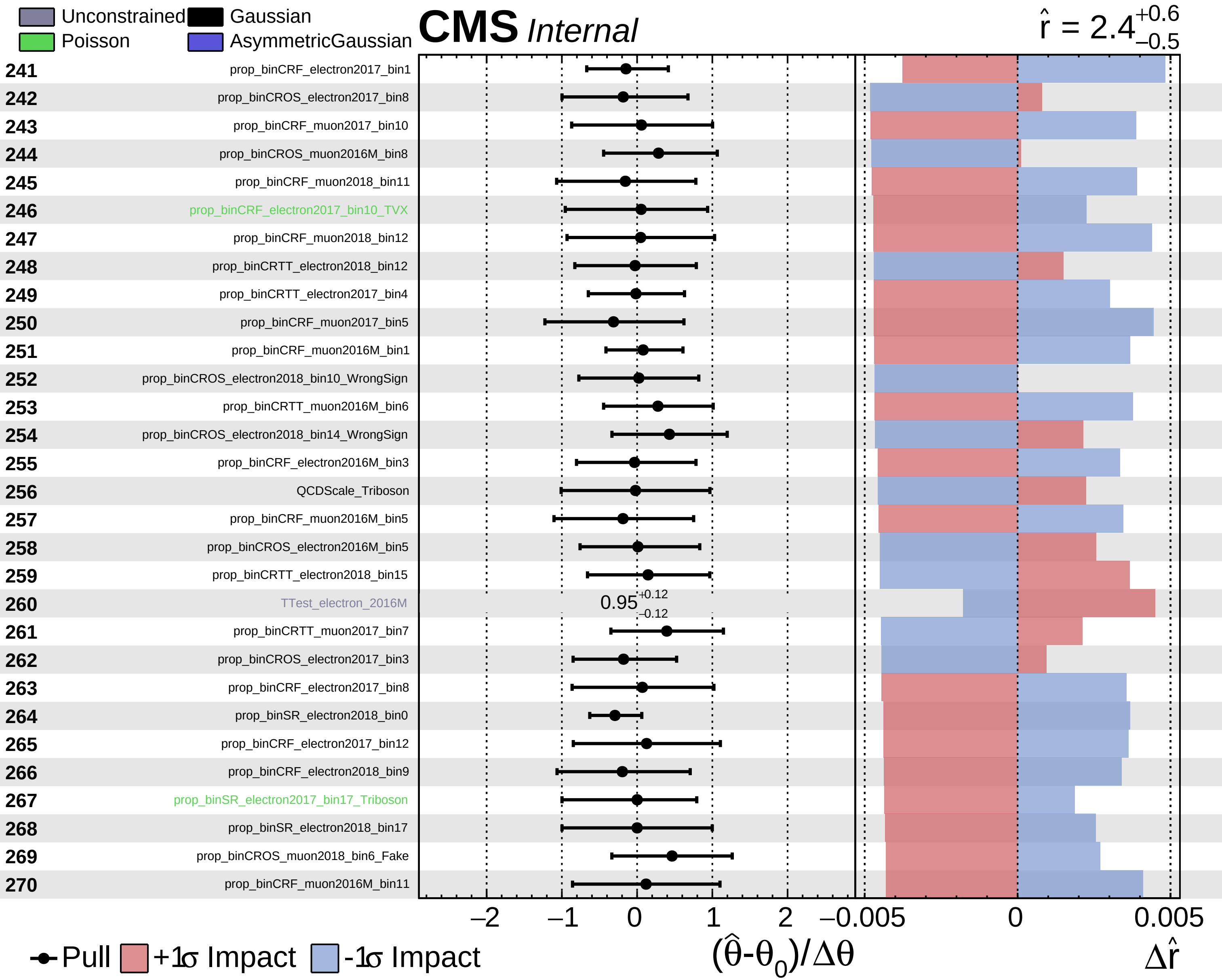


Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 2.4^{+0.6}_{-0.5}$
151 prop_binCROS_electron2018_bin5
152 prop_binCROS_muon2018_bin11
153 prop_binCRF_muon2018_bin6
154 TTest_electron_2018
$0.95^{+0.07}_{-0.07}$
155 prop_binCROS_muon2017_bin4_WrongSign
156 prop_binCRF_muon2017_bin3
157 prop_binCROS_electron2016M_bin6_WrongSign
158 prop_binSR_electron2016M_bin1
159 prop_binCRF_electron2018_bin7
160 prop_binSR_muon2018_bin10
161 prop_binSR_muon2018_bin14
162 prop_binCRF_muon2018_bin9
163 prop_binCROS_electron2016M_bin9_WrongSign
164 prop_binCROS_muon2018_bin8
165 prop_binCRTT_muon2017_bin11
166 prop_binCRTT_muon2017_bin1
167 prop_binCRF_muon2017_bin2
168 prop_binCROS_electron2018_bin7
169 prop_binCROS_muon2017_bin7
170 prop_binCRF_electron2017_bin4
171 prop_binCROS_electron2016M_bin15_WrongSign
172 prop_binSR_electron2016M_bin12
173 prop_binCROS_electron2016M_bin0
174 prop_binSR_muon2018_bin15
175 prop_binCROS_muon2017_bin5
176 prop_binCRF_electron2017_bin10_WpWpJJ_EWK
177 prop_binCRTT_muon2017_bin4
178 prop_binCRF_muon2016M_bin3
179 prop_binSR_muon2018_bin16
180 prop_binCRF_muon2016M_bin6
−2 −1 0 1 2
−0.005 0 0.005
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$

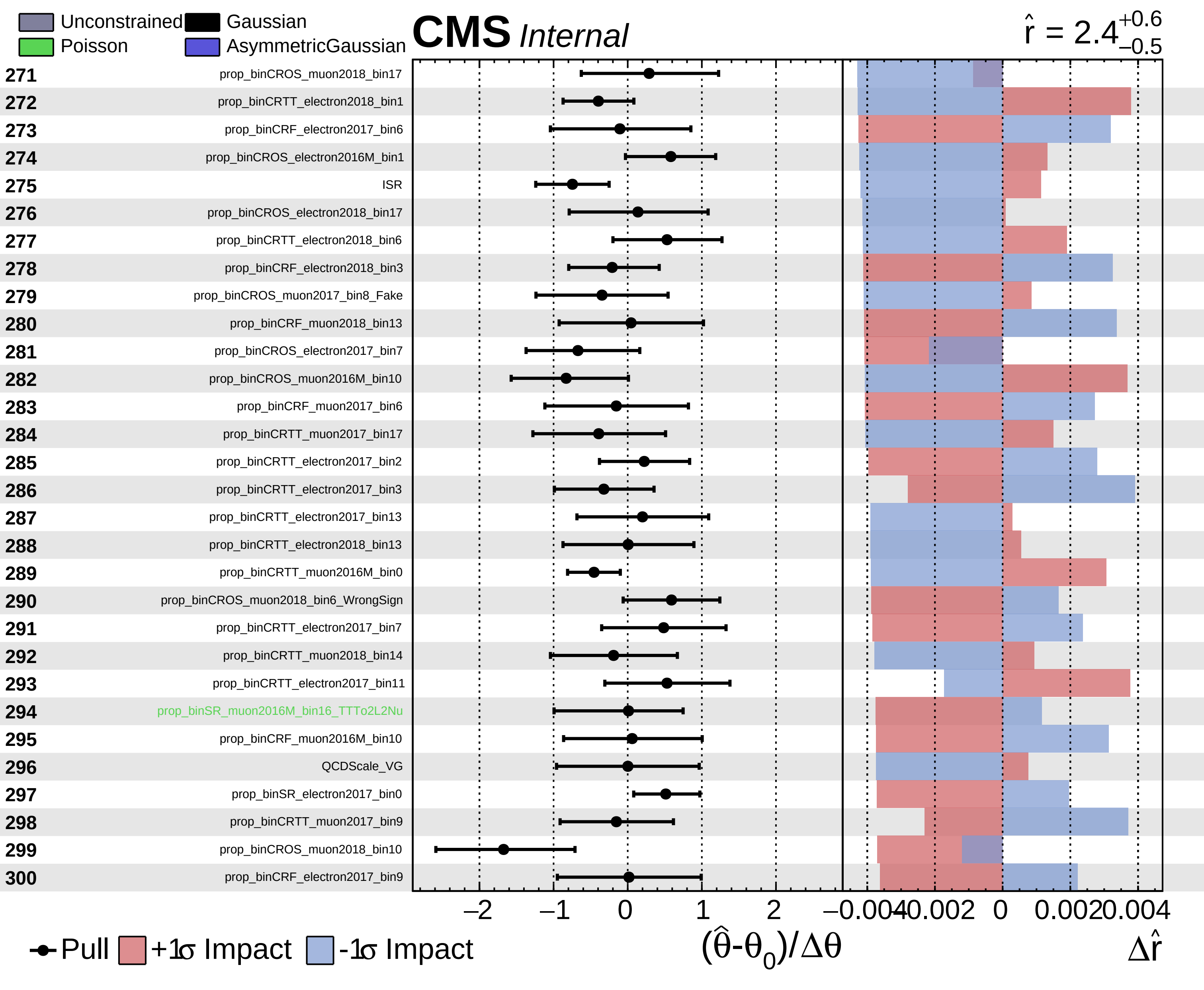


Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 2.4^{+0.6}_{-0.5}$
181 prop_binSR_electron2016M_bin3
182 prop_binSR_muon2017_bin11
183 prop_binSR_electron2018_bin15
184 prop_binSR_muon2018_bin17
185 prop_binCRF_electron2017_bin2
186 prop_binCRF_electron2017_bin5
187 prop_binSR_muon2017_bin15
188 prop_binCROS_electron2017_bin5
189 QCDScale_WrongSign
190 prop_binCRF_electron2018_bin4
191 prop_binCRF_muon2017_bin4
192 prop_binSR_electron2017_bin13_WZ
193 prop_binCROS_electron2017_bin2
194 prop_binCRTT_electron2018_bin7
195 prop_binCROS_electron2018_bin11
196 FSR
197 prop_binCRF_muon2018_bin7
198 prop_binSR_electron2018_bin14
199 prop_binCRF_muon2016M_bin7
200 prop_binSR_muon2016M_bin13
201 prop_binSR_electron2016M_bin16_TTTo2L2Nu
202 prop_binCRTT_electron2017_bin8
203 prop_binCROS_muon2016M_bin0
204 prop_binSR_muon2016M_bin12
205 prop_binCRTT_electron2018_bin8
206 prop_binCRTT_muon2018_bin7
207 prop_binSR_electron2016M_bin2_Fake
208 prop_binCRTT_electron2018_bin2
209 prop_binSR_muon2017_bin0
210 prop_binSR_muon2017_bin12
−2 −1 0 1 2
−0.005 0 0.005
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$

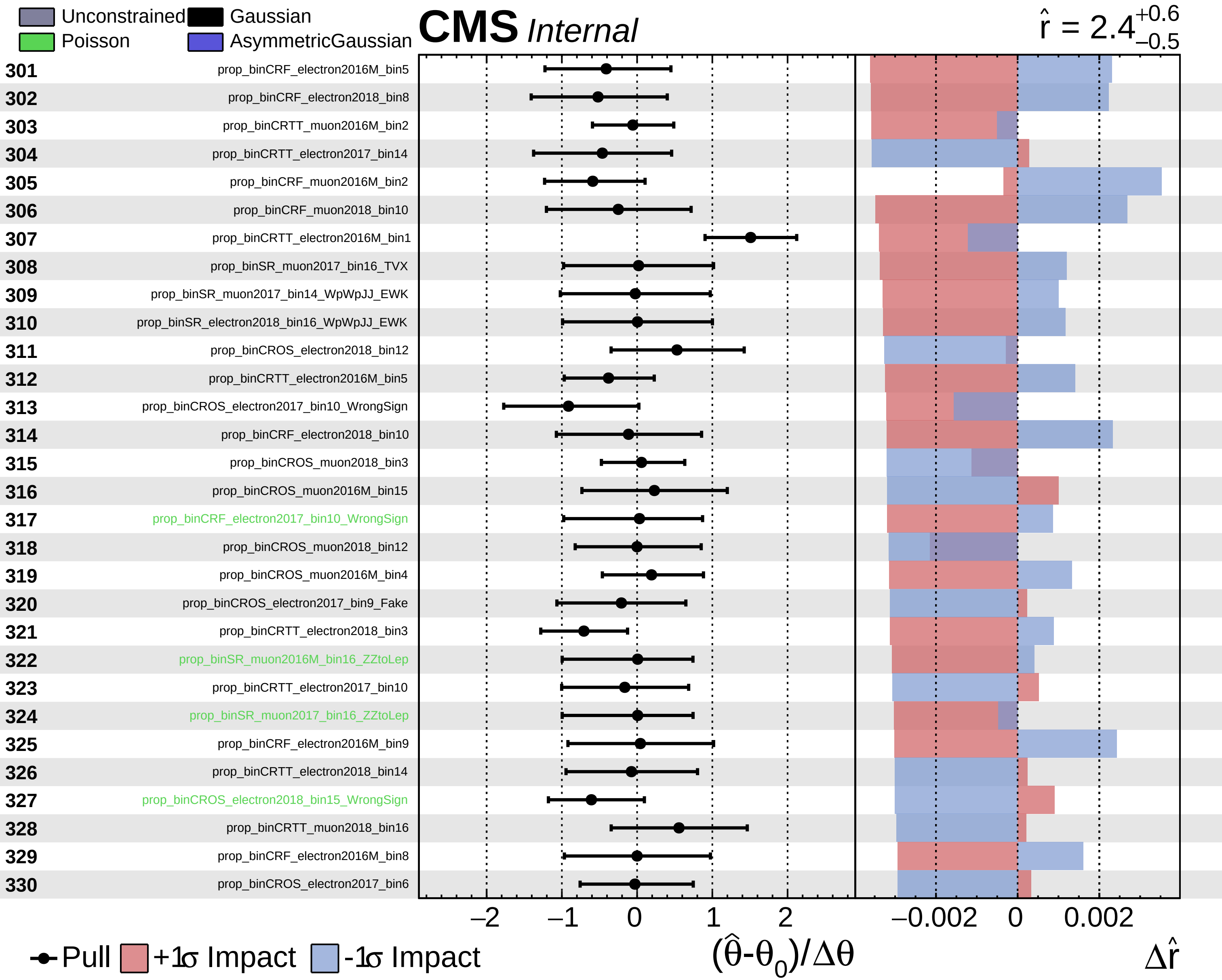

Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 2.4^{+0.6}_{-0.5}$
211 prop_binCRF_muon2016M_bin4
212 prop_binCROS_muon2017_bin4_Fake
213 prop_binSR_muon2016M_bin11
214 prop_binCRTT_muon2018_bin3
215 prop_binCRF_muon2016M_bin0
216 prop_binCROS_electron2017_bin1
217 prop_binCRF_muon2017_bin8
218 prop_binCRF_electron2016M_bin4
219 prop_binSR_electron2016M_bin16_WZ
220 prop_binCRF_electron2017_bin3
221 prop_binCRTT_muon2018_bin10
222 prop_binCRF_electron2018_bin6
223 prop_binSR_electron2017_bin11
224 prop_binCRF_muon2017_bin7
225 prop_binCROS_muon2017_bin12_WrongSign
226 prop_binCRTT_muon2018_bin17
227 prop_binCRF_electron2018_bin2
228 prop_binCRTT_muon2017_bin13
229 prop_binSR_muon2017_bin10
230 prop_binCRTT_electron2017_bin5
231 prop_binCRTT_muon2016M_bin4
232 prop_binCRF_muon2016M_bin13
233 prop_binSR_muon2016M_bin0
234 prop_binCROS_muon2016M_bin5
235 prop_binCRTT_muon2017_bin5
236 prop_binCRTT_muon2018_bin5
237 prop_binCRF_electron2018_bin5
238 prop_binCRF_electron2016M_bin2
239 prop_binCRF_electron2018_bin1
240 prop_binSR_electron2017_bin13_Fake
−2
−1
0
1
2
−0.005
0
0.005
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$



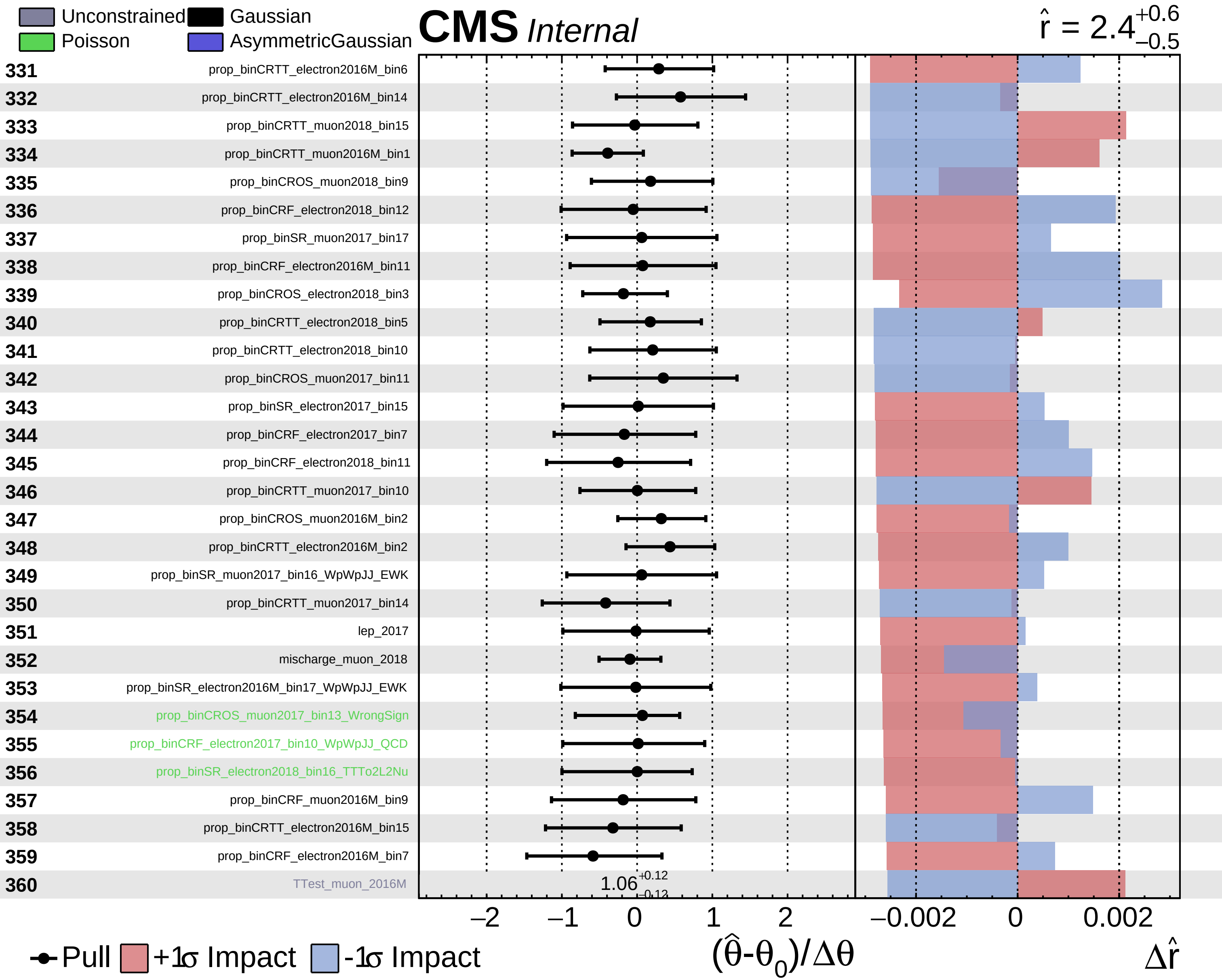
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 2.4^{+0.6}_{-0.5}$
241 prop_binCRF_electron2017_bin1
242 prop_binCROS_electron2017_bin8
243 prop_binCRF_muon2017_bin10
244 prop_binCROS_muon2016M_bin8
245 prop_binCRF_muon2018_bin11
246 prop_binCRF_electron2017_bin10_TVX
247 prop_binCRF_muon2018_bin12
248 prop_binCRTT_electron2018_bin12
249 prop_binCRTT_electron2017_bin4
250 prop_binCRF_muon2017_bin5
251 prop_binCRF_muon2016M_bin1
252 prop_binCROS_electron2018_bin10_WrongSign
253 prop_binCRTT_muon2016M_bin6
254 prop_binCROS_electron2018_bin14_WrongSign
255 prop_binCRF_electron2016M_bin3
256 QCDScale_Triboson
257 prop_binCRF_muon2016M_bin5
258 prop_binCROS_electron2016M_bin5
259 prop_binCRTT_electron2018_bin15
260 TTest_electron_2016M
$0.95^{+0.12}_{-0.12}$
261 prop_binCRTT_muon2017_bin7
262 prop_binCROS_electron2017_bin3
263 prop_binCRF_electron2017_bin8
264 prop_binSR_electron2018_bin0
265 prop_binCRF_electron2017_bin12
266 prop_binCRF_electron2018_bin9
267 prop_binSR_electron2017_bin17_Triboson
268 prop_binSR_electron2018_bin17
269 prop_binCROS_muon2018_bin6_Fake
270 prop_binCRF_muon2016M_bin11
−2
−1
0
1
2
−0.005
0
0.005
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$



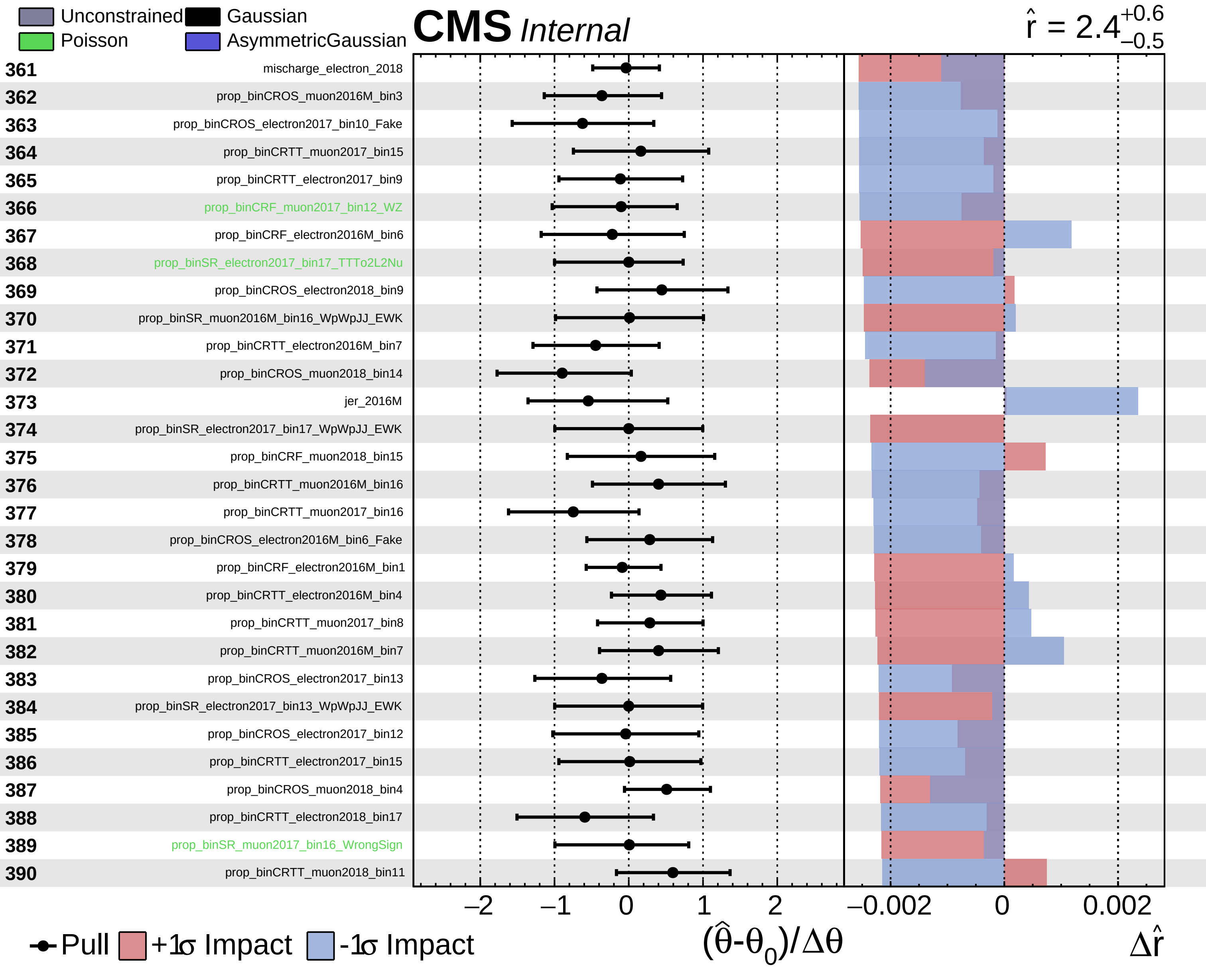
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 2.4^{+0.6}_{-0.5}$
271 prop_binCROS_muon2018_bin17
272 prop_binCRTT_electron2018_bin1
273 prop_binCRF_electron2017_bin6
274 prop_binCROS_electron2016M_bin1
275 ISR
276 prop_binCROS_electron2018_bin17
277 prop_binCRTT_electron2018_bin6
278 prop_binCRF_electron2018_bin3
279 prop_binCROS_muon2017_bin8_Fake
280 prop_binCRF_muon2018_bin13
281 prop_binCROS_electron2017_bin7
282 prop_binCROS_muon2016M_bin10
283 prop_binCRF_muon2017_bin6
284 prop_binCRTT_muon2017_bin17
285 prop_binCRTT_electron2017_bin2
286 prop_binCRTT_electron2017_bin3
287 prop_binCRTT_electron2017_bin13
288 prop_binCRTT_electron2018_bin13
289 prop_binCRTT_muon2016M_bin0
290 prop_binCROS_muon2018_bin6_WrongSign
291 prop_binCRTT_electron2017_bin7
292 prop_binCRTT_muon2018_bin14
293 prop_binCRTT_electron2017_bin11
294 prop_binSR_muon2016M_bin16_TTTo2L2Nu
295 prop_binCRF_muon2016M_bin10
296 QCDScale_VG
297 prop_binSR_electron2017_bin0
298 prop_binCRTT_muon2017_bin9
299 prop_binCROS_muon2018_bin10
300 prop_binCRF_electron2017_bin9
−2 −1 0 1 2
−0.004 −0.002 0 0.002 0.004
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$



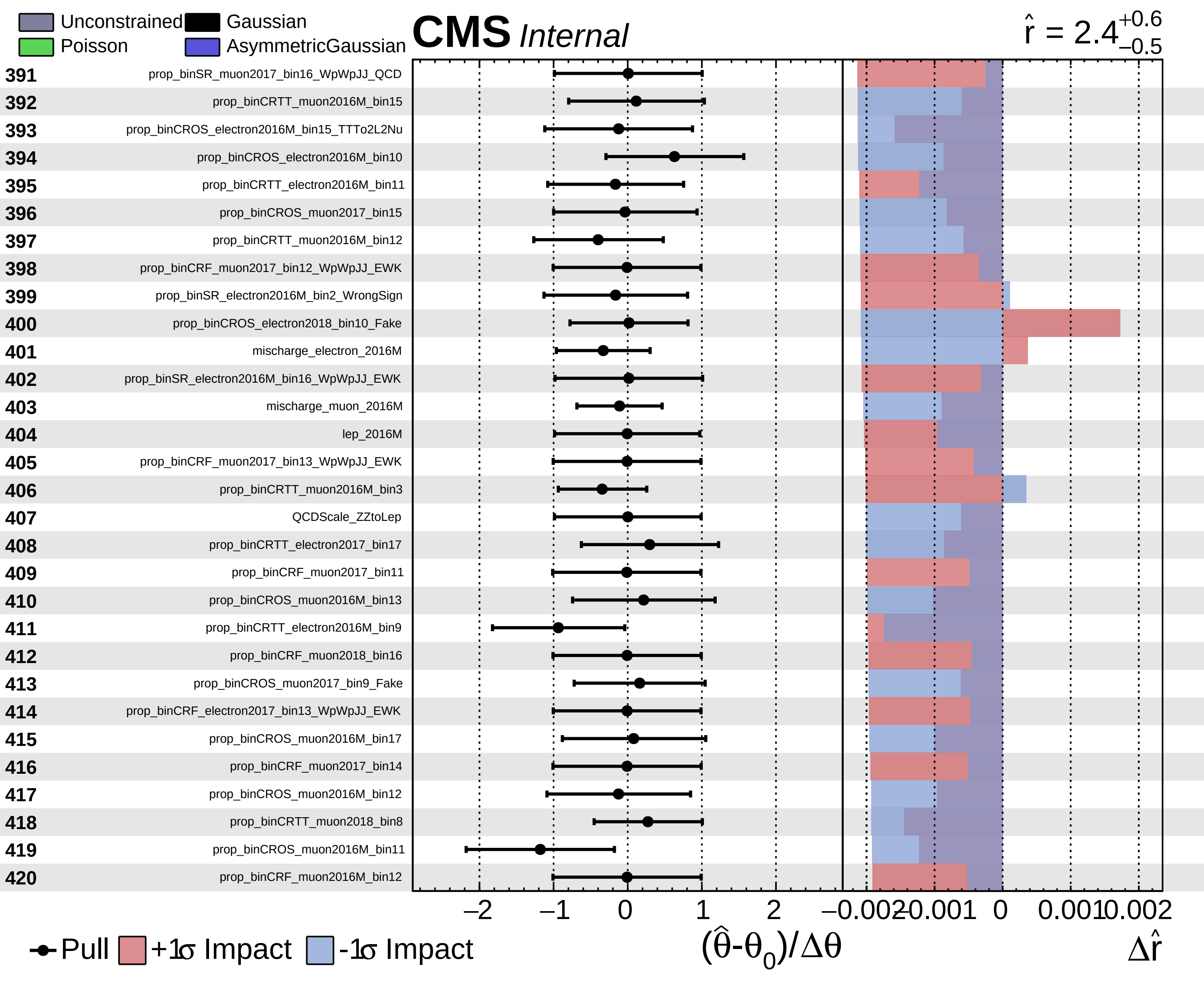
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 2.4^{+0.6}_{-0.5}$
301 prop_binCRF_electron2016M_bin5
302 prop_binCRF_electron2018_bin8
303 prop_binCRTT_muon2016M_bin2
304 prop_binCRTT_electron2017_bin14
305 prop_binCRF_muon2016M_bin2
306 prop_binCRF_muon2018_bin10
307 prop_binCRTT_electron2016M_bin1
308 prop_binSR_muon2017_bin16_TVX
309 prop_binSR_muon2017_bin14_WpWpJJ_EWK
310 prop_binSR_electron2018_bin16_WpWpJJ_EWK
311 prop_binCROS_electron2018_bin12
312 prop_binCRTT_electron2016M_bin5
313 prop_binCROS_electron2017_bin10_WrongSign
314 prop_binCRF_electron2018_bin10
315 prop_binCROS_muon2018_bin3
316 prop_binCROS_muon2016M_bin15
317 prop_binCRF_electron2017_bin10_WrongSign
318 prop_binCROS_muon2018_bin12
319 prop_binCROS_muon2016M_bin4
320 prop_binCROS_electron2017_bin9_Fake
321 prop_binCRTT_electron2018_bin3
322 prop_binSR_muon2016M_bin16_ZZtoLep
323 prop_binCRTT_electron2017_bin10
324 prop_binSR_muon2017_bin16_ZZtoLep
325 prop_binCRF_electron2016M_bin9
326 prop_binCRTT_electron2018_bin14
327 prop_binCROS_electron2018_bin15_WrongSign
328 prop_binCRTT_muon2018_bin16
329 prop_binCRF_electron2016M_bin8
330 prop_binCROS_electron2017_bin6
−2
−1
0
1
2
−0.002
0
0.002
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$



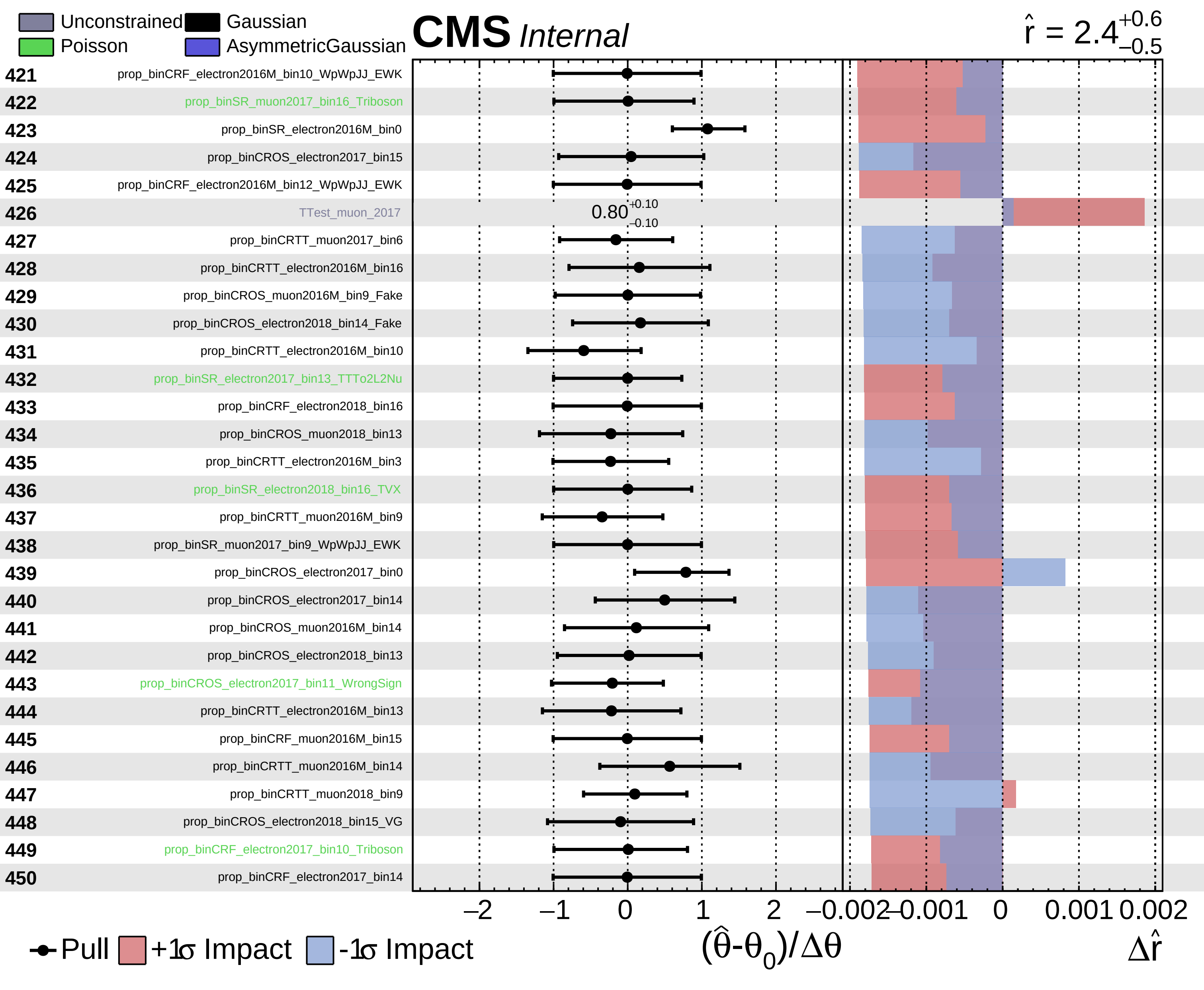
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 2.4^{+0.6}_{-0.5}$
331 prop_binCRTT_electron2016M_bin6
332 prop_binCRTT_electron2016M_bin14
333 prop_binCRTT_muon2018_bin15
334 prop_binCRTT_muon2016M_bin1
335 prop_binCROS_muon2018_bin9
336 prop_binCRF_electron2018_bin12
337 prop_binSR_muon2017_bin17
338 prop_binCRF_electron2016M_bin11
339 prop_binCROS_electron2018_bin3
340 prop_binCRTT_electron2018_bin5
341 prop_binCRTT_electron2018_bin10
342 prop_binCROS_muon2017_bin11
343 prop_binSR_electron2017_bin15
344 prop_binCRF_electron2017_bin7
345 prop_binCRF_electron2018_bin11
346 prop_binCRTT_muon2017_bin10
347 prop_binCROS_muon2016M_bin2
348 prop_binCRTT_electron2016M_bin2
349 prop_binSR_muon2017_bin16_WpWpJJ_EWK
350 prop_binCRTT_muon2017_bin14
351 lep_2017
352 mischarge_muon_2018
353 prop_binSR_electron2016M_bin17_WpWpJJ_EWK
354 prop_binCROS_muon2017_bin13_WrongSign
355 prop_binCRF_electron2017_bin10_WpWpJJ_QCD
356 prop_binSR_electron2018_bin16_TTTo2L2Nu
357 prop_binCRF_muon2016M_bin9
358 prop_binCRTT_electron2016M_bin15
359 prop_binCRF_electron2016M_bin7
360 TTest_muon_2016M
$1.06^{+0.12}_{-0.12}$
−2 −1 0 1 2
−0.002 0 0.002
Pull +1σ Impact -1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$



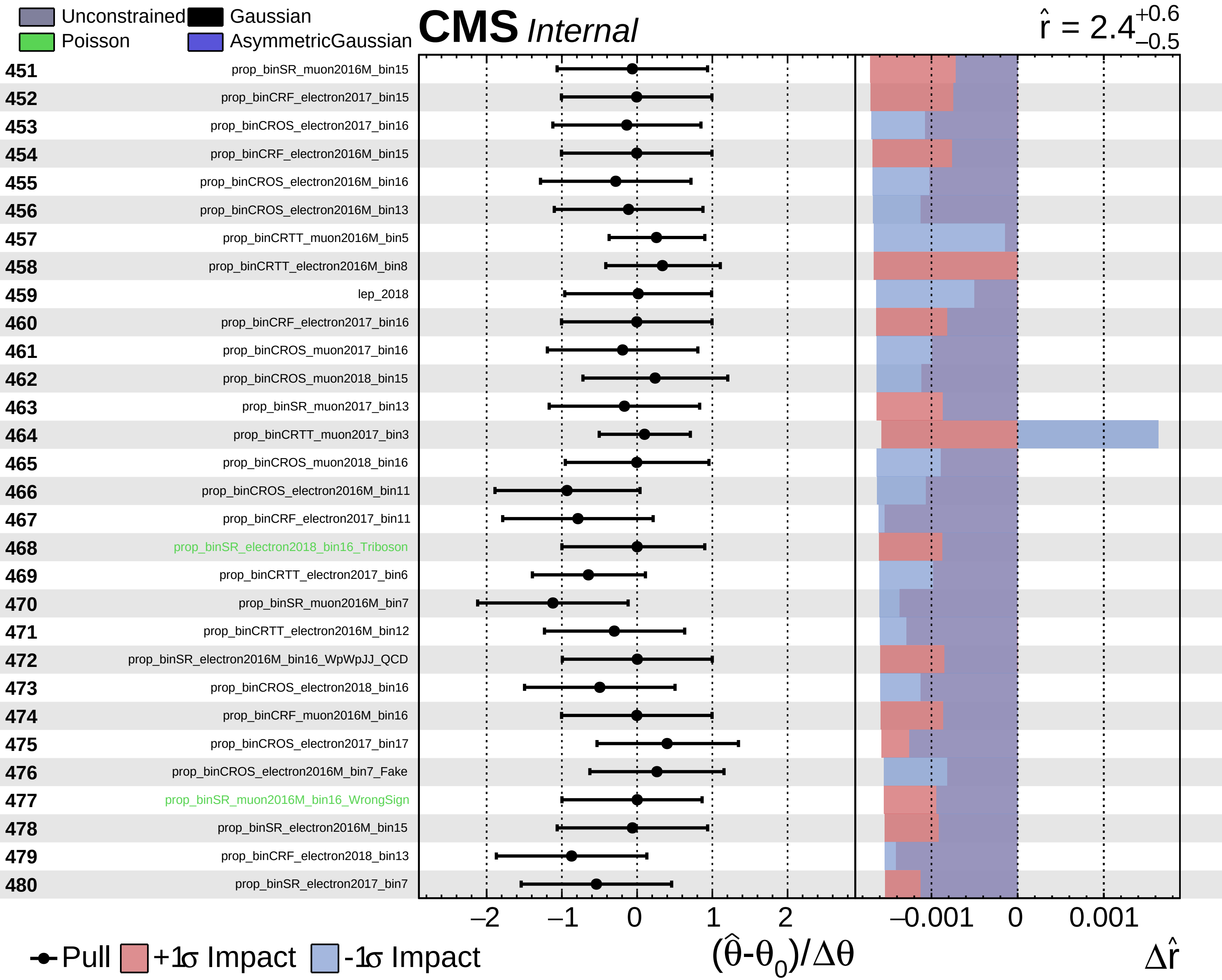
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 2.4^{+0.6}_{-0.5}$
361 mischarge_electron_2018
362 prop_binCROS_muon2016M_bin3
363 prop_binCROS_electron2017_bin10_Fake
364 prop_binCRTT_muon2017_bin15
365 prop_binCRTT_electron2017_bin9
366 prop_binCRF_muon2017_bin12_WZ
367 prop_binCRF_electron2016M_bin6
368 prop_binSR_electron2017_bin17_TTTo2L2Nu
369 prop_binCROS_electron2018_bin9
370 prop_binSR_muon2016M_bin16_WpWpJJ_EWK
371 prop_binCRTT_electron2016M_bin7
372 prop_binCROS_muon2018_bin14
373 jer_2016M
374 prop_binSR_electron2017_bin17_WpWpJJ_EWK
375 prop_binCRF_muon2018_bin15
376 prop_binCRTT_muon2016M_bin16
377 prop_binCRTT_muon2017_bin16
378 prop_binCROS_electron2016M_bin6_Fake
379 prop_binCRF_electron2016M_bin1
380 prop_binCRTT_electron2016M_bin4
381 prop_binCRTT_muon2017_bin8
382 prop_binCRTT_muon2016M_bin7
383 prop_binCROS_electron2017_bin13
384 prop_binSR_electron2017_bin13_WpWpJJ_EWK
385 prop_binCROS_electron2017_bin12
386 prop_binCRTT_electron2017_bin15
387 prop_binCROS_muon2018_bin4
388 prop_binCRTT_electron2018_bin17
389 prop_binSR_muon2017_bin16_WrongSign
390 prop_binCRTT_muon2018_bin11
−2
−1
0
1
2
−0.002
0
0.002
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$



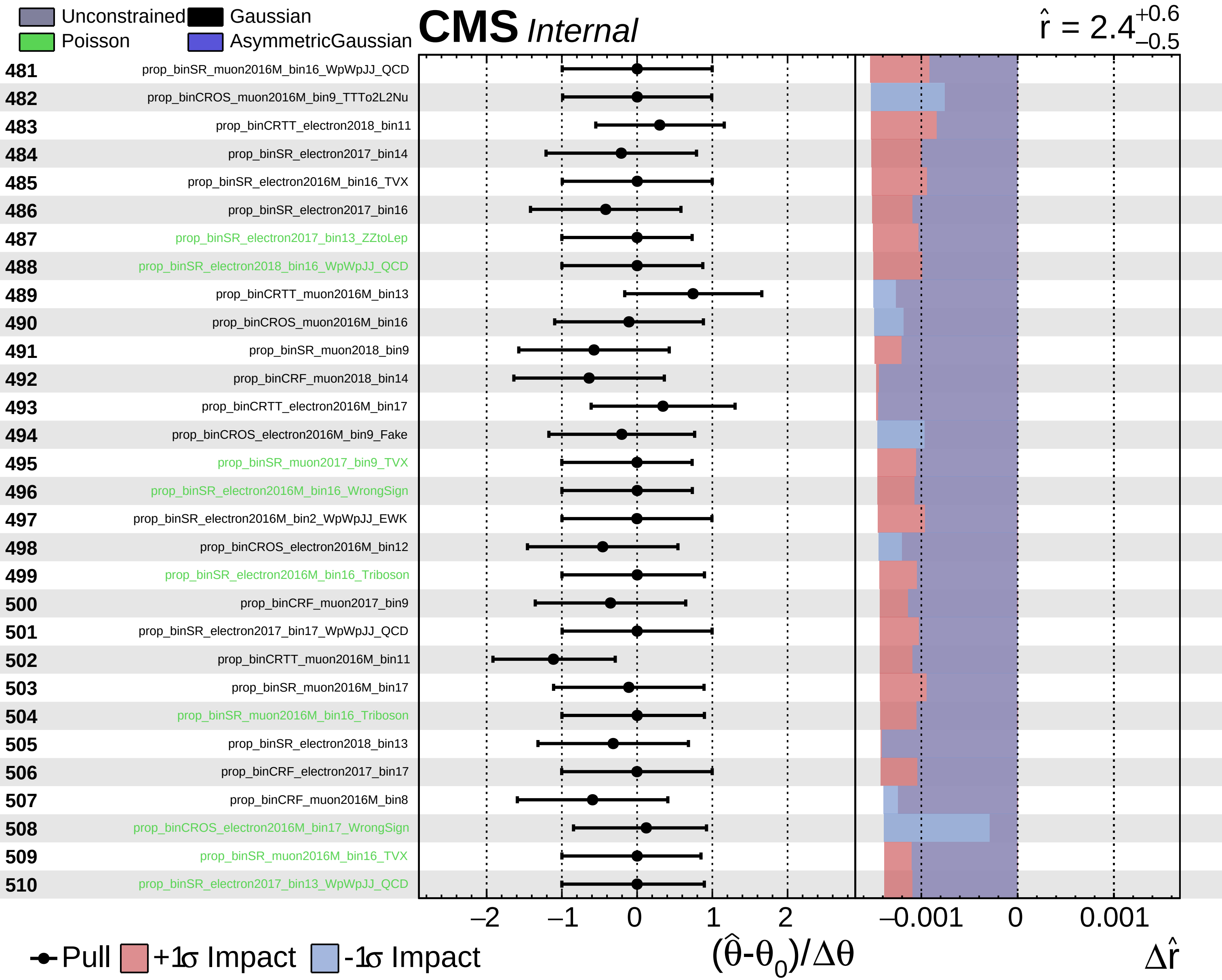
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 2.4^{+0.6}_{-0.5}$
391 prop_binSR_muon2017_bin16_WpWpJJ_QCD
392 prop_binCRTT_muon2016M_bin15
393 prop_binCROS_electron2016M_bin15_TTTo2L2Nu
394 prop_binCROS_electron2016M_bin10
395 prop_binCRTT_electron2016M_bin11
396 prop_binCROS_muon2017_bin15
397 prop_binCRTT_muon2016M_bin12
398 prop_binCRF_muon2017_bin12_WpWpJJ_EWK
399 prop_binSR_electron2016M_bin2_WrongSign
400 prop_binCROS_electron2018_bin10_Fake
401 mischarge_electron_2016M
402 prop_binSR_electron2016M_bin16_WpWpJJ_EWK
403 mischarge_muon_2016M
404 lep_2016M
405 prop_binCRF_muon2017_bin13_WpWpJJ_EWK
406 prop_binCRTT_muon2016M_bin3
407 QCDScale_ZZtoLep
408 prop_binCRTT_electron2017_bin17
409 prop_binCRF_muon2017_bin11
410 prop_binCROS_muon2016M_bin13
411 prop_binCRTT_electron2016M_bin9
412 prop_binCRF_muon2018_bin16
413 prop_binCROS_muon2017_bin9_Fake
414 prop_binCRF_electron2017_bin13_WpWpJJ_EWK
415 prop_binCROS_muon2016M_bin17
416 prop_binCRF_muon2017_bin14
417 prop_binCROS_muon2016M_bin12
418 prop_binCRTT_muon2018_bin8
419 prop_binCROS_muon2016M_bin11
420 prop_binCRF_muon2016M_bin12
−2 −1 0 1 2
−0.002 −0.001 0 0.001 0.002
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$

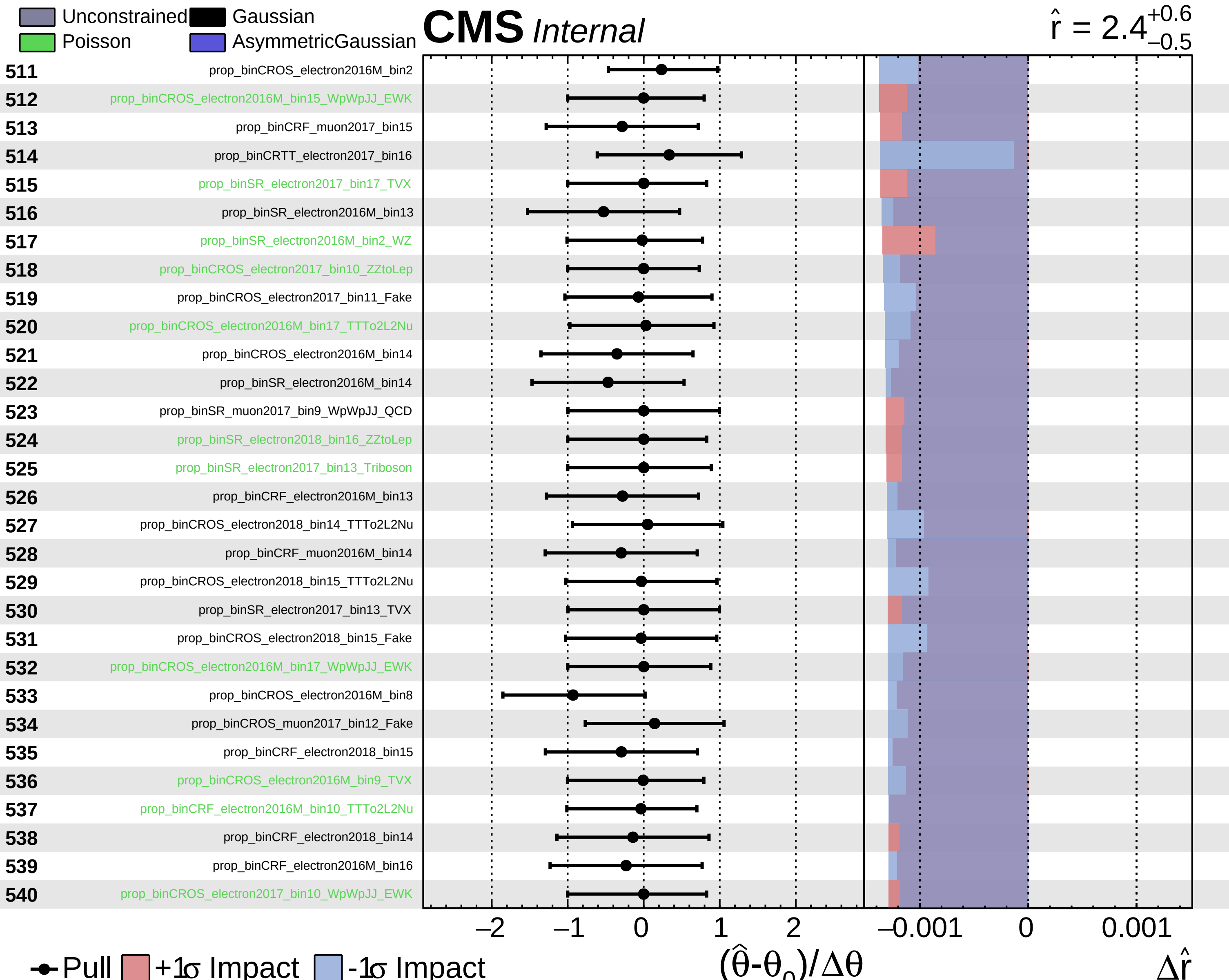


Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 2.4^{+0.6}_{-0.5}$
421 prop_binCRF_electron2016M_bin10_WpWpJJ_EWK
422 prop_binSR_muon2017_bin16_Triboson
423 prop_binSR_electron2016M_bin0
424 prop_binCROS_electron2017_bin15
425 prop_binCRF_electron2016M_bin12_WpWpJJ_EWK
426 TTest_muon_2017
$0.80^{+0.10}_{-0.10}$
427 prop_binCRTT_muon2017_bin6
428 prop_binCRTT_electron2016M_bin16
429 prop_binCROS_muon2016M_bin9_Fake
430 prop_binCROS_electron2018_bin14_Fake
431 prop_binCRTT_electron2016M_bin10
432 prop_binSR_electron2017_bin13_TTTo2L2Nu
433 prop_binCRF_electron2018_bin16
434 prop_binCROS_muon2018_bin13
435 prop_binCRTT_electron2016M_bin3
436 prop_binSR_electron2018_bin16_TVX
437 prop_binCRTT_muon2016M_bin9
438 prop_binSR_muon2017_bin9_WpWpJJ_EWK
439 prop_binCROS_electron2017_bin0
440 prop_binCROS_electron2017_bin14
441 prop_binCROS_muon2016M_bin14
442 prop_binCROS_electron2018_bin13
443 prop_binCROS_electron2017_bin11_WrongSign
444 prop_binCRTT_electron2016M_bin13
445 prop_binCRF_muon2016M_bin15
446 prop_binCRTT_muon2016M_bin14
447 prop_binCRTT_muon2018_bin9
448 prop_binCROS_electron2018_bin15_VG
449 prop_binCRF_electron2017_bin10_Triboson
450 prop_binCRF_electron2017_bin14
−2 −1 0 1 2
−0.002 −0.001 0 0.001 0.002
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$

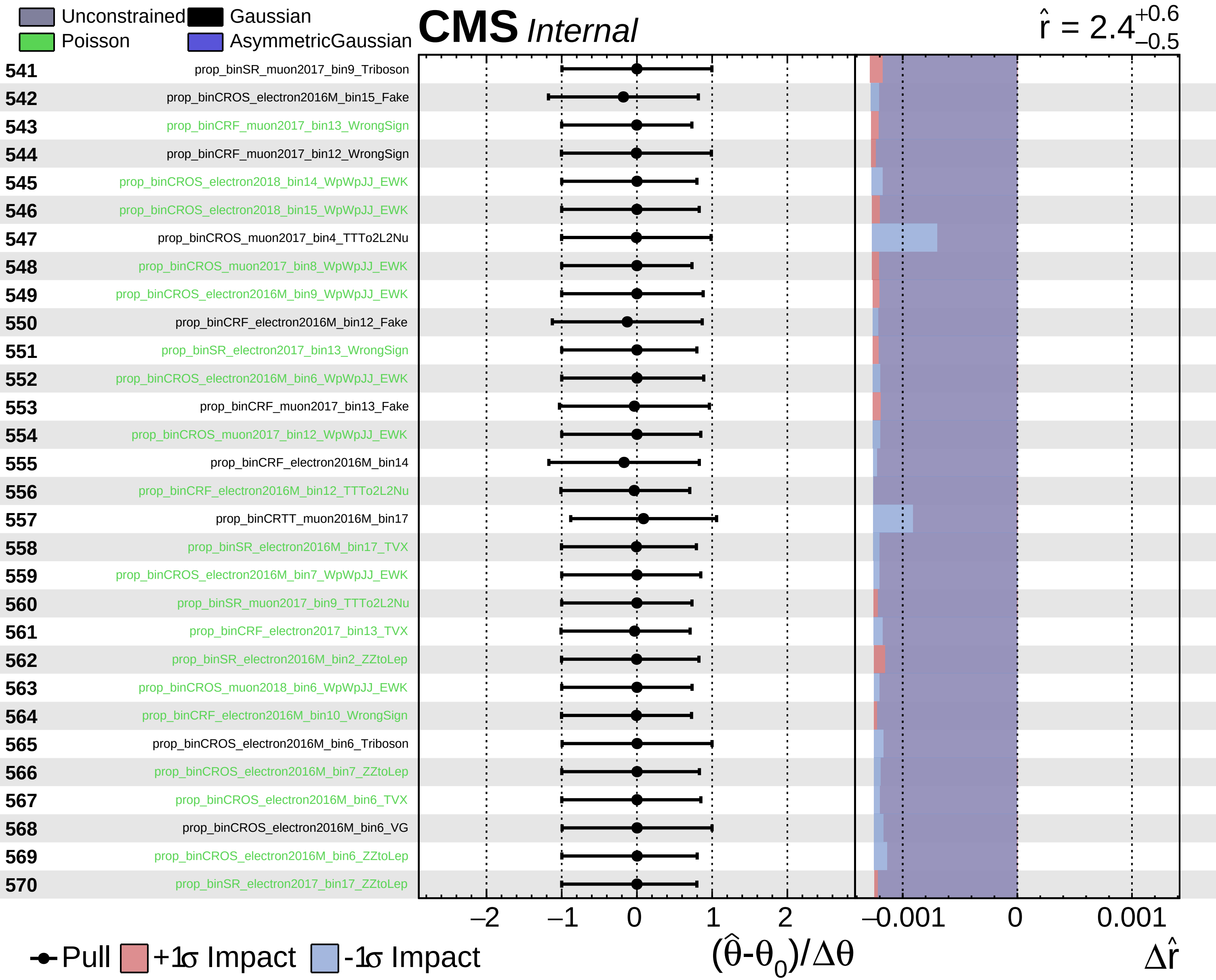


Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 2.4^{+0.6}_{-0.5}$
451 prop_binSR_muon2016M_bin15
452 prop_binCRF_electron2017_bin15
453 prop_binCROS_electron2017_bin16
454 prop_binCRF_electron2016M_bin15
455 prop_binCROS_electron2016M_bin16
456 prop_binCROS_electron2016M_bin13
457 prop_binCRTT_muon2016M_bin5
458 prop_binCRTT_electron2016M_bin8
459 lep_2018
460 prop_binCRF_electron2017_bin16
461 prop_binCROS_muon2017_bin16
462 prop_binCROS_muon2018_bin15
463 prop_binSR_muon2017_bin13
464 prop_binCRTT_muon2017_bin3
465 prop_binCROS_muon2018_bin16
466 prop_binCROS_electron2016M_bin11
467 prop_binCRF_electron2017_bin11
468 prop_binSR_electron2018_bin16_Triboson
469 prop_binCRTT_electron2017_bin6
470 prop_binSR_muon2016M_bin7
471 prop_binCRTT_electron2016M_bin12
472 prop_binSR_electron2016M_bin16_WpWpJJ_QCD
473 prop_binCROS_electron2018_bin16
474 prop_binCRF_muon2016M_bin16
475 prop_binCROS_electron2017_bin17
476 prop_binCROS_electron2016M_bin7_Fake
477 prop_binSR_muon2016M_bin16_WrongSign
478 prop_binSR_electron2016M_bin15
479 prop_binCRF_electron2018_bin13
480 prop_binSR_electron2017_bin7
−2 −1 0 1 2
−0.001 0 0.001
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$

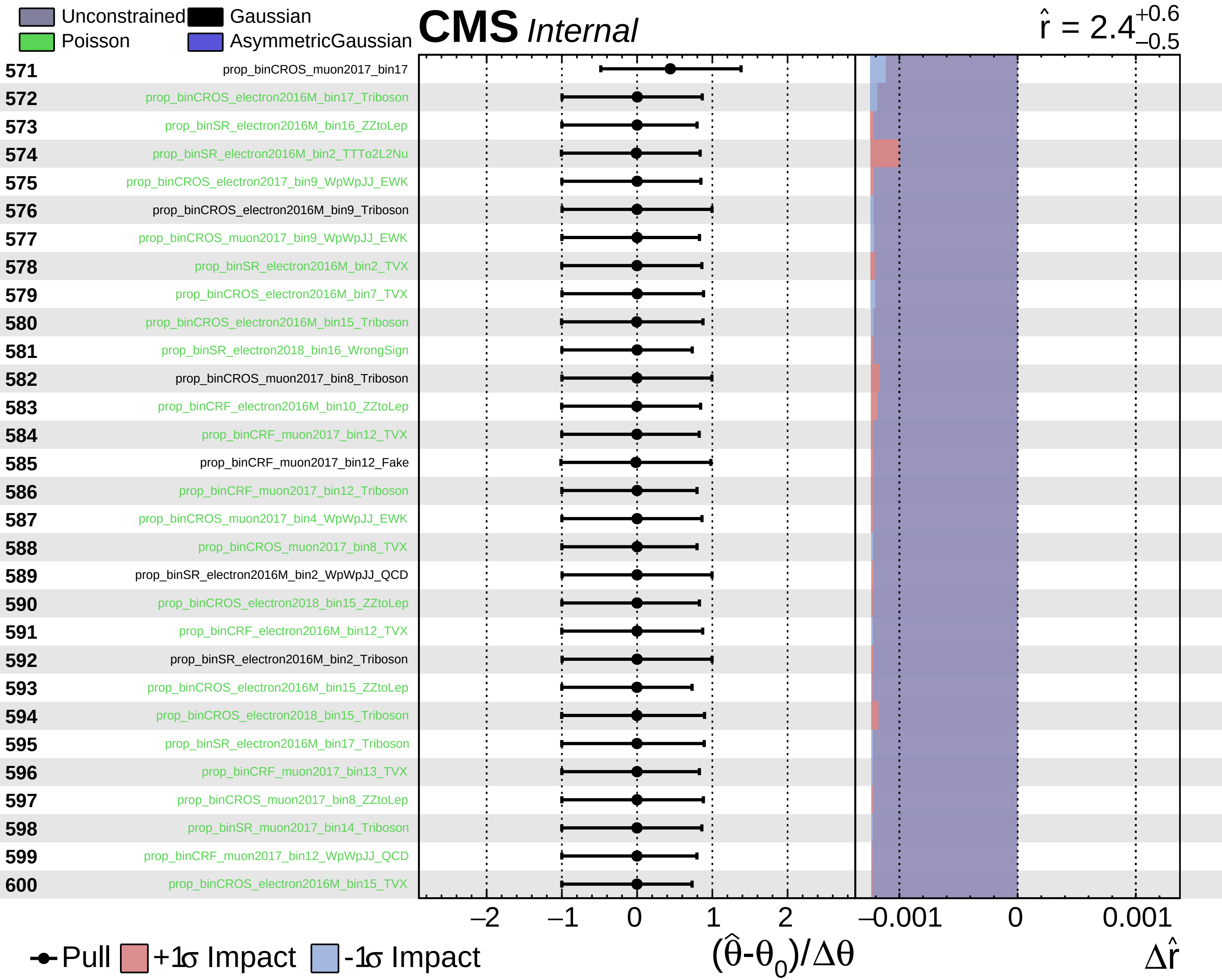

Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 2.4^{+0.6}_{-0.5}$
481 prop_binSR_muon2016M_bin16_WpWpJJ_QCD
482 prop_binCROS_muon2016M_bin9_TTTo2L2Nu
483 prop_binCRTT_electron2018_bin11
484 prop_binSR_electron2017_bin14
485 prop_binSR_electron2016M_bin16_TVX
486 prop_binSR_electron2017_bin16
487 prop_binSR_electron2017_bin13_ZZtoLep
488 prop_binSR_electron2018_bin16_WpWpJJ_QCD
489 prop_binCRTT_muon2016M_bin13
490 prop_binCROS_muon2016M_bin16
491 prop_binSR_muon2018_bin9
492 prop_binCRF_muon2018_bin14
493 prop_binCRTT_electron2016M_bin17
494 prop_binCROS_electron2016M_bin9_Fake
495 prop_binSR_muon2017_bin9_TVX
496 prop_binSR_electron2016M_bin16_WrongSign
497 prop_binSR_electron2016M_bin2_WpWpJJ_EWK
498 prop_binCROS_electron2016M_bin12
499 prop_binSR_electron2016M_bin16_Triboson
500 prop_binCRF_muon2017_bin9
501 prop_binSR_electron2017_bin17_WpWpJJ_QCD
502 prop_binCRTT_muon2016M_bin11
503 prop_binSR_muon2016M_bin17
504 prop_binSR_muon2016M_bin16_Triboson
505 prop_binSR_electron2018_bin13
506 prop_binCRF_electron2017_bin17
507 prop_binCRF_muon2016M_bin8
508 prop_binCROS_electron2016M_bin17_WrongSign
509 prop_binSR_muon2016M_bin16_TVX
510 prop_binSR_electron2017_bin13_WpWpJJ_QCD
−2 −1 0 1 2
−0.001 0 0.001
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$



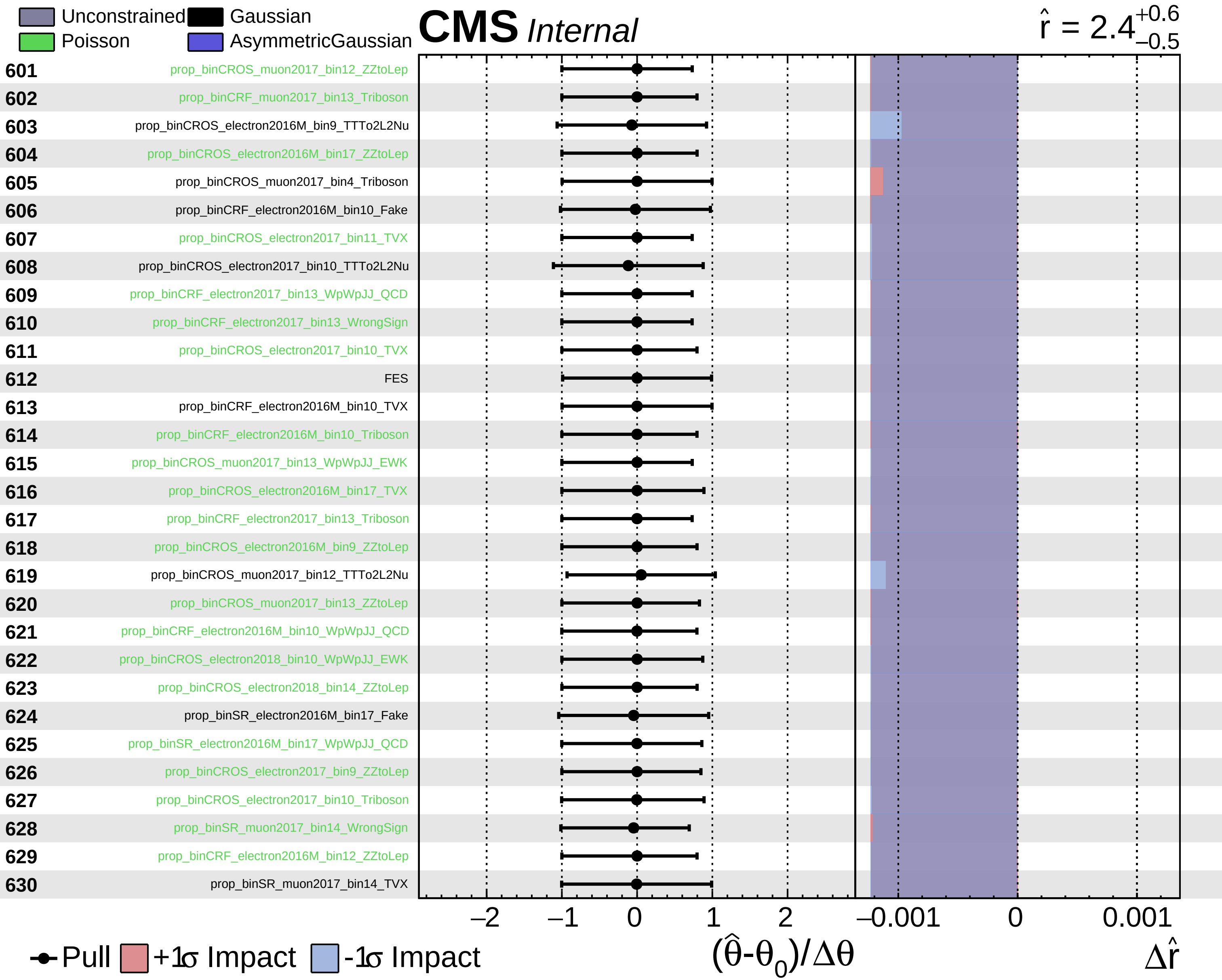
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 2.4^{+0.6}_{-0.5}$
511 prop_binCROS_electron2016M_bin2
512 prop_binCROS_electron2016M_bin15_WpWpJJ_EWK
513 prop_binCRF_muon2017_bin15
514 prop_binCRTT_electron2017_bin16
515 prop_binSR_electron2017_bin17_TVX
516 prop_binSR_electron2016M_bin13
517 prop_binSR_electron2016M_bin2_WZ
518 prop_binCROS_electron2017_bin10_ZZtoLep
519 prop_binCROS_electron2017_bin11_Fake
520 prop_binCROS_electron2016M_bin17_TTTo2L2Nu
521 prop_binCROS_electron2016M_bin14
522 prop_binSR_electron2016M_bin14
523 prop_binSR_muon2017_bin9_WpWpJJ_QCD
524 prop_binSR_electron2018_bin16_ZZtoLep
525 prop_binSR_electron2017_bin13_Triboson
526 prop_binCRF_electron2016M_bin13
527 prop_binCROS_electron2018_bin14_TTTo2L2Nu
528 prop_binCRF_muon2016M_bin14
529 prop_binCROS_electron2018_bin15_TTTo2L2Nu
530 prop_binSR_electron2017_bin13_TVX
531 prop_binCROS_electron2018_bin15_Fake
532 prop_binCROS_electron2016M_bin17_WpWpJJ_EWK
533 prop_binCROS_electron2016M_bin8
534 prop_binCROS_muon2017_bin12_Fake
535 prop_binCRF_electron2018_bin15
536 prop_binCROS_electron2016M_bin9_TVX
537 prop_binCRF_electron2016M_bin10_TTTo2L2Nu
538 prop_binCRF_electron2018_bin14
539 prop_binCRF_electron2016M_bin16
540 prop_binCROS_electron2017_bin10_WpWpJJ_EWK
−2 −1 0 1 2
−0.001 0 0.001
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$



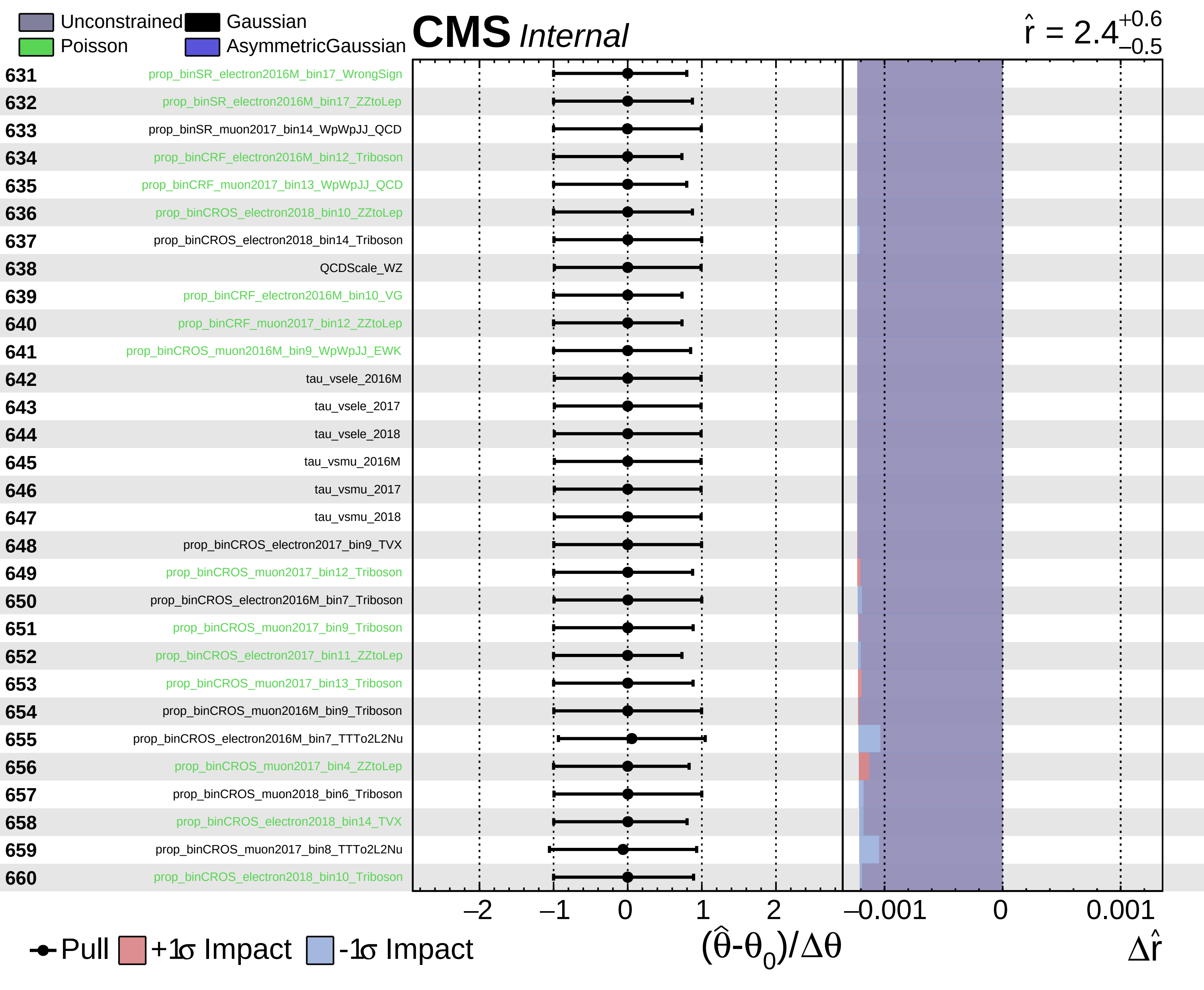
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 2.4^{+0.6}_{-0.5}$
541 prop_binSR_muon2017_bin9_Triboson
542 prop_binCROS_electron2016M_bin15_Fake
543 prop_binCRF_muon2017_bin13_WrongSign
544 prop_binCRF_muon2017_bin12_WrongSign
545 prop_binCROS_electron2018_bin14_WpWpJJ_EWK
546 prop_binCROS_electron2018_bin15_WpWpJJ_EWK
547 prop_binCROS_muon2017_bin4_TTTo2L2Nu
548 prop_binCROS_muon2017_bin8_WpWpJJ_EWK
549 prop_binCROS_electron2016M_bin9_WpWpJJ_EWK
550 prop_binCRF_electron2016M_bin12_Fake
551 prop_binSR_electron2017_bin13_WrongSign
552 prop_binCROS_electron2016M_bin6_WpWpJJ_EWK
553 prop_binCRF_muon2017_bin13_Fake
554 prop_binCROS_muon2017_bin12_WpWpJJ_EWK
555 prop_binCRF_electron2016M_bin14
556 prop_binCRF_electron2016M_bin12_TTTo2L2Nu
557 prop_binCRTT_muon2016M_bin17
558 prop_binSR_electron2016M_bin17_TVX
559 prop_binCROS_electron2016M_bin7_WpWpJJ_EWK
560 prop_binSR_muon2017_bin9_TTTo2L2Nu
561 prop_binCRF_electron2017_bin13_TVX
562 prop_binSR_electron2016M_bin2_ZZtoLep
563 prop_binCROS_muon2018_bin6_WpWpJJ_EWK
564 prop_binCRF_electron2016M_bin10_WrongSign
565 prop_binCROS_electron2016M_bin6_Triboson
566 prop_binCROS_electron2016M_bin7_ZZtoLep
567 prop_binCROS_electron2016M_bin6_TVX
568 prop_binCROS_electron2016M_bin6_VG
569 prop_binCROS_electron2016M_bin6_ZZtoLep
570 prop_binSR_electron2017_bin17_ZZtoLep
−2
−1
0
1
2
−0.001
0
0.001
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$



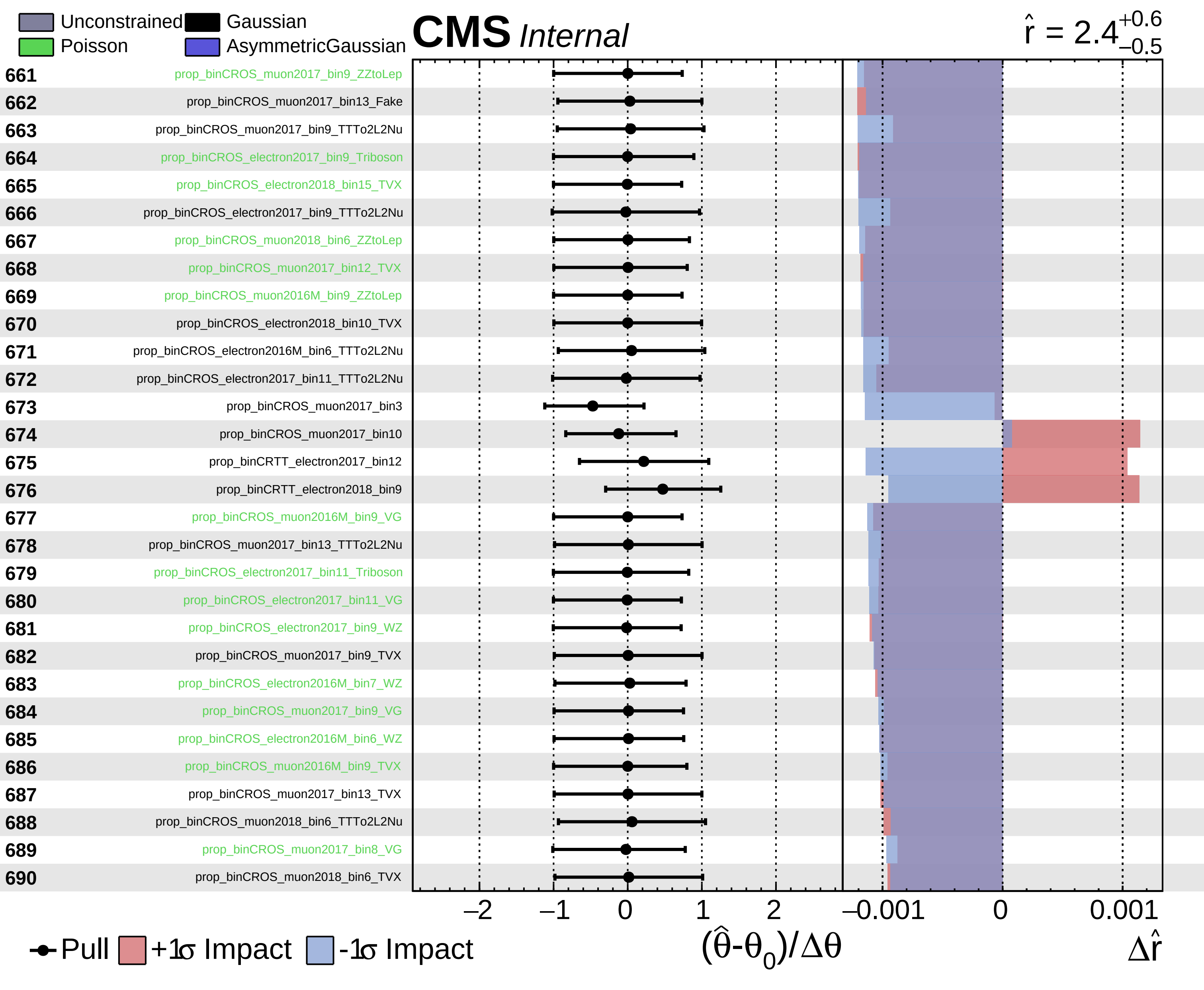
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 2.4^{+0.6}_{-0.5}$
571 prop_binCROS_muon2017_bin17
572 prop_binCROS_electron2016M_bin17_Triboson
573 prop_binSR_electron2016M_bin16_ZZtoLep
574 prop_binSR_electron2016M_bin2_TTTo2L2Nu
575 prop_binCROS_electron2017_bin9_WpWpJJ_EWK
576 prop_binCROS_electron2016M_bin9_Triboson
577 prop_binCROS_muon2017_bin9_WpWpJJ_EWK
578 prop_binSR_electron2016M_bin2_TVX
579 prop_binCROS_electron2016M_bin7_TVX
580 prop_binCROS_electron2016M_bin15_Triboson
581 prop_binSR_electron2018_bin16_WrongSign
582 prop_binCROS_muon2017_bin8_Triboson
583 prop_binCRF_electron2016M_bin10_ZZtoLep
584 prop_binCRF_muon2017_bin12_TVX
585 prop_binCRF_muon2017_bin12_Fake
586 prop_binCRF_muon2017_bin12_Triboson
587 prop_binCROS_muon2017_bin4_WpWpJJ_EWK
588 prop_binCROS_muon2017_bin8_TVX
589 prop_binSR_electron2016M_bin2_WpWpJJ_QCD
590 prop_binCROS_electron2018_bin15_ZZtoLep
591 prop_binCRF_electron2016M_bin12_TVX
592 prop_binSR_electron2016M_bin2_Triboson
593 prop_binCROS_electron2016M_bin15_ZZtoLep
594 prop_binCROS_electron2018_bin15_Triboson
595 prop_binSR_electron2016M_bin17_Triboson
596 prop_binCRF_muon2017_bin13_TVX
597 prop_binCROS_muon2017_bin8_ZZtoLep
598 prop_binSR_muon2017_bin14_Triboson
599 prop_binCRF_muon2017_bin12_WpWpJJ_QCD
600 prop_binCROS_electron2016M_bin15_TVX
−2
−1
0
1
2
−0.001
0
0.001
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$



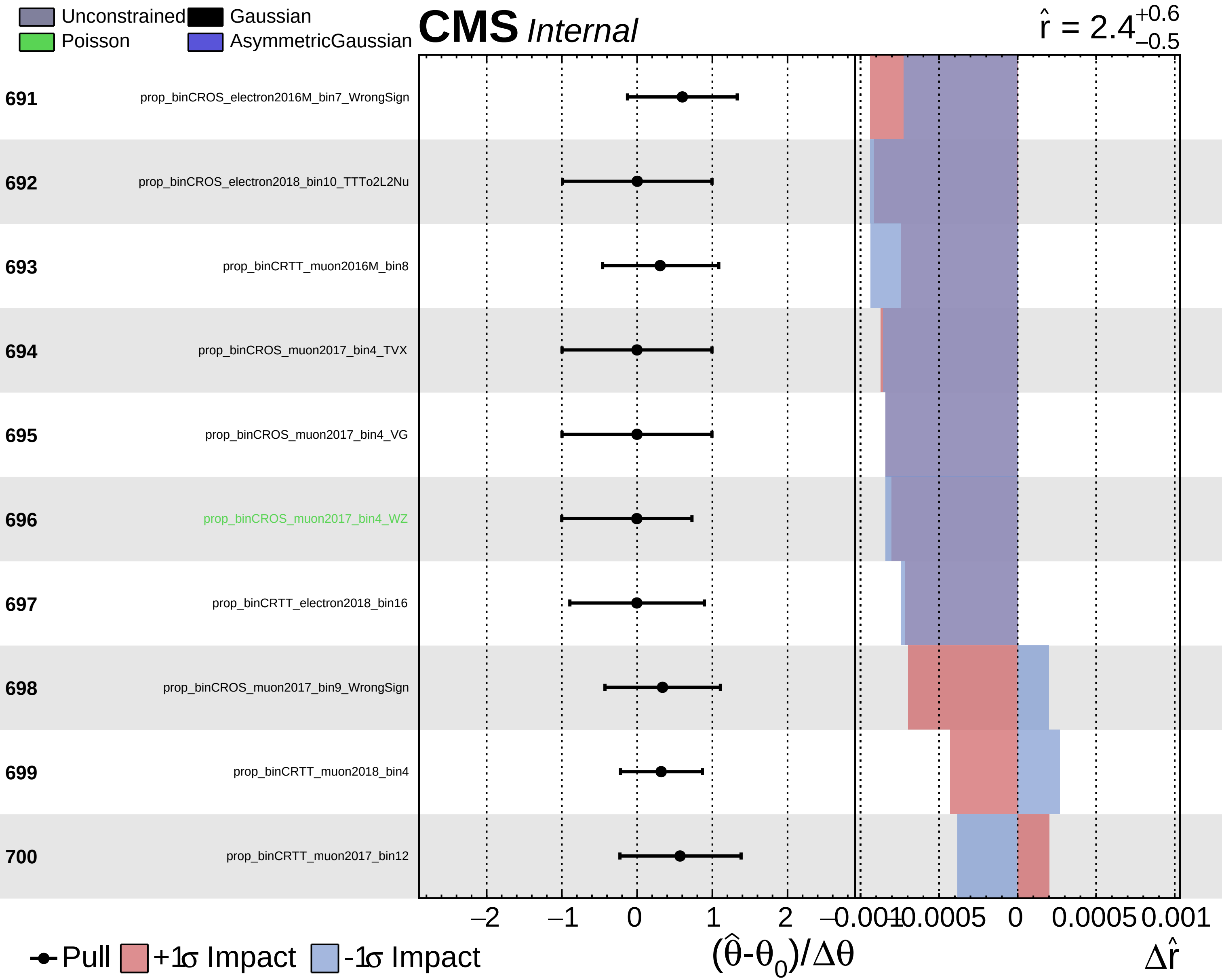
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 2.4^{+0.6}_{-0.5}$
601 prop_binCROS_muon2017_bin12_ZZtoLep
602 prop_binCRF_muon2017_bin13_Triboson
603 prop_binCROS_electron2016M_bin9_TTTo2L2Nu
604 prop_binCROS_electron2016M_bin17_ZZtoLep
605 prop_binCROS_muon2017_bin4_Triboson
606 prop_binCRF_electron2016M_bin10_Fake
607 prop_binCROS_electron2017_bin11_TVX
608 prop_binCROS_electron2017_bin10_TTTo2L2Nu
609 prop_binCRF_electron2017_bin13_WpWpJJ_QCD
610 prop_binCRF_electron2017_bin13_WrongSign
611 prop_binCROS_electron2017_bin10_TVX
612 FES
613 prop_binCRF_electron2016M_bin10_TVX
614 prop_binCRF_electron2016M_bin10_Triboson
615 prop_binCROS_muon2017_bin13_WpWpJJ_EWK
616 prop_binCROS_electron2016M_bin17_TVX
617 prop_binCRF_electron2017_bin13_Triboson
618 prop_binCROS_electron2016M_bin9_ZZtoLep
619 prop_binCROS_muon2017_bin12_TTTo2L2Nu
620 prop_binCROS_muon2017_bin13_ZZtoLep
621 prop_binCRF_electron2016M_bin10_WpWpJJ_QCD
622 prop_binCROS_electron2018_bin10_WpWpJJ_EWK
623 prop_binCROS_electron2018_bin14_ZZtoLep
624 prop_binSR_electron2016M_bin17_Fake
625 prop_binSR_electron2016M_bin17_WpWpJJ_QCD
626 prop_binCROS_electron2017_bin9_ZZtoLep
627 prop_binCROS_electron2017_bin10_Triboson
628 prop_binSR_muon2017_bin14_WrongSign
629 prop_binCRF_electron2016M_bin12_ZZtoLep
630 prop_binSR_muon2017_bin14_TVX
−2
−1
0
1
2
−0.001
0
0.001
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$



Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 2.4^{+0.6}_{-0.5}$
631 prop_binSR_electron2016M_bin17_WrongSign
632 prop_binSR_electron2016M_bin17_ZZtoLep
633 prop_binSR_muon2017_bin14_WpWpJJ_QCD
634 prop_binCRF_electron2016M_bin12_Triboson
635 prop_binCRF_muon2017_bin13_WpWpJJ_QCD
636 prop_binCROS_electron2018_bin10_ZZtoLep
637 prop_binCROS_electron2018_bin14_Triboson
638 QCDScale_WZ
639 prop_binCRF_electron2016M_bin10_VG
640 prop_binCRF_muon2017_bin12_ZZtoLep
641 prop_binCROS_muon2016M_bin9_WpWpJJ_EWK
642 tau_vsele_2016M
643 tau_vsele_2017
644 tau_vsele_2018
645 tau_vsmu_2016M
646 tau_vsmu_2017
647 tau_vsmu_2018
648 prop_binCROS_electron2017_bin9_TVX
649 prop_binCROS_muon2017_bin12_Triboson
650 prop_binCROS_electron2016M_bin7_Triboson
651 prop_binCROS_muon2017_bin9_Triboson
652 prop_binCROS_electron2017_bin11_ZZtoLep
653 prop_binCROS_muon2017_bin13_Triboson
654 prop_binCROS_muon2016M_bin9_Triboson
655 prop_binCROS_electron2016M_bin7_TTTo2L2Nu
656 prop_binCROS_muon2017_bin4_ZZtoLep
657 prop_binCROS_muon2018_bin6_Triboson
658 prop_binCROS_electron2018_bin14_TVX
659 prop_binCROS_muon2017_bin8_TTTo2L2Nu
660 prop_binCROS_electron2018_bin10_Triboson
−2
−1
0
1
2
−0.001
0
0.001
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$



Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 2.4^{+0.6}_{-0.5}$
661 prop_binCROS_muon2017_bin9_ZZtoLep
662 prop_binCROS_muon2017_bin13_Fake
663 prop_binCROS_muon2017_bin9_TTTo2L2Nu
664 prop_binCROS_electron2017_bin9_Triboson
665 prop_binCROS_electron2018_bin15_TVX
666 prop_binCROS_electron2017_bin9_TTTo2L2Nu
667 prop_binCROS_muon2018_bin6_ZZtoLep
668 prop_binCROS_muon2017_bin12_TVX
669 prop_binCROS_muon2016M_bin9_ZZtoLep
670 prop_binCROS_electron2018_bin10_TVX
671 prop_binCROS_electron2016M_bin6_TTTo2L2Nu
672 prop_binCROS_electron2017_bin11_TTTo2L2Nu
673 prop_binCROS_muon2017_bin3
674 prop_binCROS_muon2017_bin10
675 prop_binCRTT_electron2017_bin12
676 prop_binCRTT_electron2018_bin9
677 prop_binCROS_muon2016M_bin9_VG
678 prop_binCROS_muon2017_bin13_TTTo2L2Nu
679 prop_binCROS_electron2017_bin11_Triboson
680 prop_binCROS_electron2017_bin11_VG
681 prop_binCROS_electron2017_bin9_WZ
682 prop_binCROS_muon2017_bin9_TVX
683 prop_binCROS_electron2016M_bin7_WZ
684 prop_binCROS_muon2017_bin9_VG
685 prop_binCROS_electron2016M_bin6_WZ
686 prop_binCROS_muon2016M_bin9_TVX
687 prop_binCROS_muon2017_bin13_TVX
688 prop_binCROS_muon2018_bin6_TTTo2L2Nu
689 prop_binCROS_muon2017_bin8_VG
690 prop_binCROS_muon2018_bin6_TVX
−2 −1 0 1 2
−0.001 0 0.001
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$



Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 2.4^{+0.6}_{-0.5}$
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692 prop_binCROS_electron2018_bin10_TTTo2L2Nu
693 prop_binCRTT_muon2016M_bin8
694 prop_binCROS_muon2017_bin4_TVX
695 prop_binCROS_muon2017_bin4_VG
696 prop_binCROS_muon2017_bin4_WZ
697 prop_binCRTT_electron2018_bin16
698 prop_binCROS_muon2017_bin9_WrongSign
699 prop_binCRTT_muon2018_bin4
700 prop_binCRTT_muon2017_bin12
−2 −1 0 1 2
−0.001 −0.0005 0 0.0005 0.001
Pull +1σ Impact -1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$